UNIT 1

DC & AC Circuits

Topics: Circuit elements (R, L, C) • Ohm's law • KCL & KVL • Series/parallel circuits • Superposition theorem • AC fundamentals • RMS, average, form & peak factor • Phasors in R, L, C • Impedance • Active, reactive & apparent power • Power factor

START HERE
Big picture: Electricity is the flow of charge. Every electrical circuit has a **source** (battery/supply), a **path** (wires) and a **load** (bulb, motor, heater). Unit I teaches the rules that tell us how much voltage and current each part gets.

1 DC CIRCUITS

1.1 Circuit Elements: R, L and C

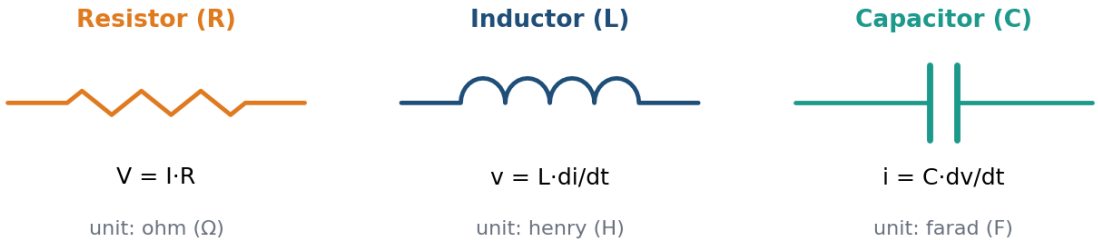


Fig 1.1 Symbols of the three passive elements

Element	What it does	Energy	Behaviour in steady DC
R Resistor	Opposes current; turns electrical energy into heat	Dissipated: $P = I^2R$	Acts as resistance R
L Inductor	Opposes *change* of current (magnetic field)	Stored: $\frac{1}{2}LI^2$ (joule)	Short circuit (plain wire)
C Capacitor	Opposes *change* of voltage (electric field)	Stored: $\frac{1}{2}CV^2$ (joule)	Open circuit (blocks DC)

DEFINITION
Resistance of a wire: $R = \rho l/A$, where ρ = resistivity (Ω·m), l = length, A = cross-section area. Longer/thinner wire → more resistance.

APPLICATIONS
R: heaters, bulbs, fuses, voltage dividers. **L:** chokes, transformers, motors, relays. **C:** filters, power-supply smoothing, fan/motor starting, tuning circuits.

1.2 Ohm’s Law and its Limitations

DEFINITION

Ohm’s Law: At constant temperature, the current through a conductor is directly proportional to the voltage across it.

$$V = I \times R \Rightarrow I = V / R \Rightarrow R = V / I$$

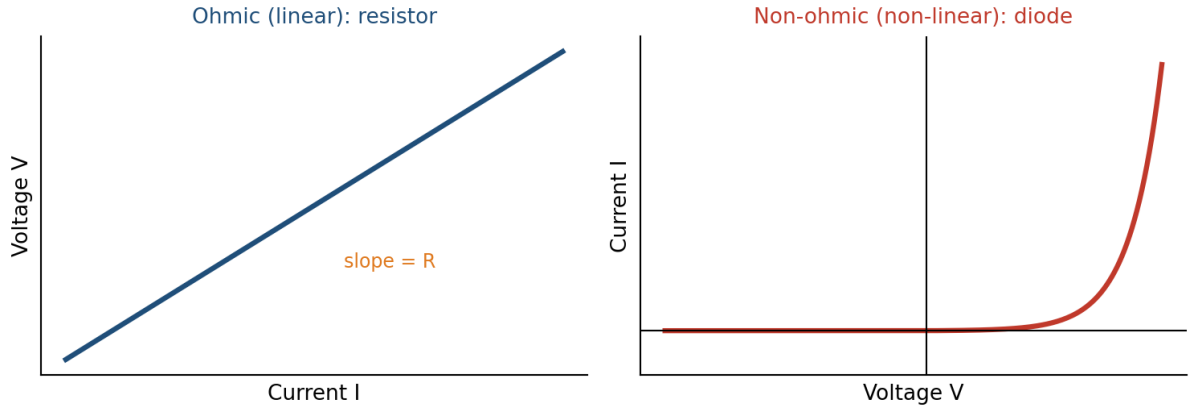


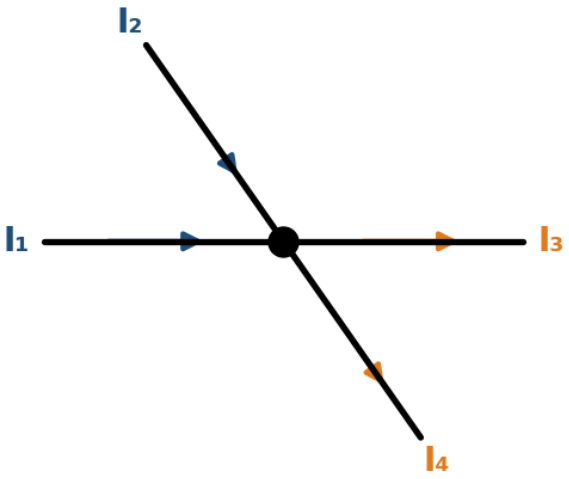
Fig 1.2 Ohmic (straight line) vs non-ohmic (curved) behaviour

Limitations of Ohm’s law

- Not valid if the **temperature** changes (resistance of metals rises with heat).
- Not applicable to **non-linear devices**: diodes, transistors, thermistors, SCRs.
- Not applicable to electrolytes, arc lamps, vacuum tubes and other **unilateral** elements.

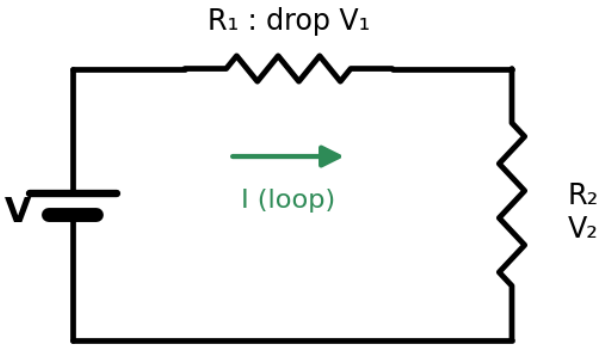
1.3 Kirchhoff’s Laws

Kirchhoff’s Current Law



KCL: $I_1 + I_2 = I_3 + I_4$
(current in = current out)

Kirchhoff’s Voltage Law



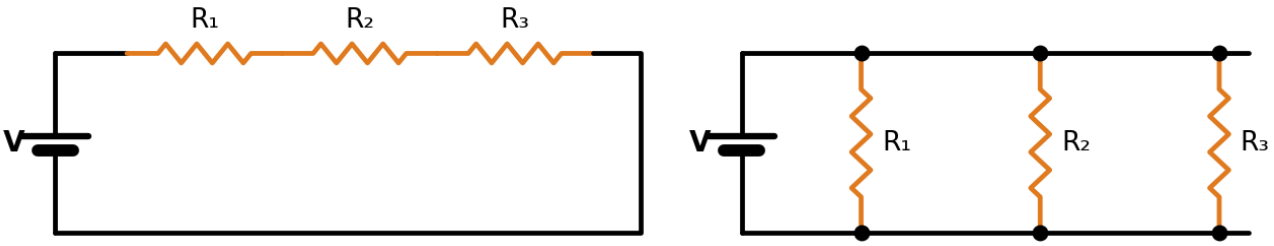
KVL: $V - V_1 - V_2 = 0$
(sum of voltages around a loop = 0)

Fig 1.3 KCL at a junction (left) and KVL around a loop (right)

	KCL (Current Law)	KVL (Voltage Law)
Statement	Algebraic sum of currents at a node = 0 (current in = current out)	Algebraic sum of voltages (rises and drops) in a closed loop = 0
Based on	Conservation of charge	Conservation of energy
Equation	$\Sigma I_{in} = \Sigma I_{out}$	$\Sigma V = 0$ (Source V = sum of drops)

EXAM TIP
Steps for KVL: (1) mark current direction in each branch, (2) choose a loop direction, (3) write drop = +IR if you go *along* the current, (4) a source is + when you pass from − to +, (5) set the sum = 0 and solve.

1.4 Series, Parallel and Series-Parallel Circuits



SERIES: same current I

R_eq = R₁ + R₂ + R₃

PARALLEL: same voltage V

1/R_eq = 1/R₁ + 1/R₂ + 1/R₃

Fig 1.4 Series and parallel connections of resistors

Property	Series	Parallel
Current	Same in all (I)	Divides: I = I₁ + I₂ + I₃
Voltage	Divides: V = V₁ + V₂ + V₃	Same across all (V)
Equivalent R	R₁ + R₂ + R₃ (larger than the biggest)	1/R₁ + 1/R₂ + 1/R₃ (smaller than the smallest)
Two resistors	—	R_{eq} = R₁R₂ / (R₁ + R₂)
Use	Fuse, switch, string lights	Household wiring (each device gets full V)

SOLVED EXAMPLE
Q1. 12 Ω, 6 Ω and 4 Ω are in parallel across 12 V. Find R_{eq} and branch currents.
$1/R_{eq}$ = 1/12 + 1/6 + 1/4 = 6/12 = 0.5 ⇒ **R_{eq} = 2 Ω**; total I = 12/2 = **6 A**. Branches: 12/12 = **1 A**, 12/6 = **2 A**, 12/4 = **3 A** (sum = 6 A ✓).
Q2 (series-parallel). 4 Ω in series with (6 Ω ∥ 12 Ω) across 24 V. 6∥12 = 4 Ω ⇒ R_{total} = 8 Ω ⇒ I = **3 A**. Voltage across the parallel part = 3 × 4 = 12 V ⇒ currents 2 A and 1 A.

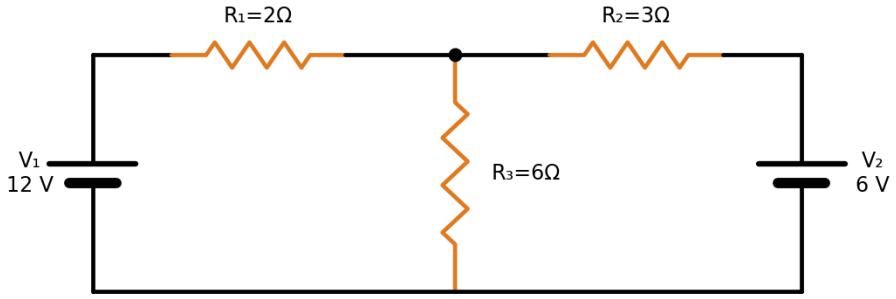
HANDY SHORTCUTS
Voltage divider (series): V₁ = V × R₁/(R₁+R₂) | **Current divider** (two parallel branches): I₁ = I × R₂/(R₁+R₂)

REMEMBER
In a **series** circuit the *largest* resistor gets the *largest* voltage; in a **parallel** circuit the *smallest* resistor carries the *largest* current. Household appliances are connected in **parallel** so that each gets full supply voltage and works independently.

1.5 Superposition Theorem

DEFINITION

In a **linear** circuit with more than one source, the current in any branch equals the **algebraic sum** of the currents produced by each source acting **alone**, with the other sources replaced by their internal resistance (**voltage source → short circuit; current source → open circuit**).



Find the current through R₃

Fig 1.5 Example circuit for superposition

SOLVED EXAMPLE

Step 1 (V_1 = 12 V only, V_2 shorted): $R_2 \| R_3 = 3\times6/9 = 2\ \Omega$; total = 2 + 2 = 4 Ω; I = 3 A; voltage across $R_3 = 3 \times 2 = 6$ V ⇒ $I_3' = 6/6 =$ **1 A**.

Step 2 (V_2 = 6 V only, V_1 shorted): $R_1 \| R_3 = 2\times6/8 = 1.5\ \Omega$; total = 3 + 1.5 = 4.5 Ω; I = 1.333 A; voltage across $R_3 = 1.333 \times 1.5 = 2$ V ⇒ $I_3'' = 2/6 =$ **0.333 A**.

Step 3 (add): $I_3 = 1 + 0.333 =$ **1.33 A** (downward). Check by node analysis: $V_A = 8$ V, 8/6 = 1.33 A ✓.

REMEMBER

Superposition works only for **linear** circuits and for current/voltage — **not for power** (power ∝ I^2, which is not linear).

2 AC CIRCUITS

2.1 AC Fundamentals

Alternating current (AC) changes its magnitude and direction periodically; the usual shape is a **sine wave**. Mains supply in India: **230 V, 50 Hz**. AC is preferred because voltage can be easily stepped up/down by transformers for efficient transmission.

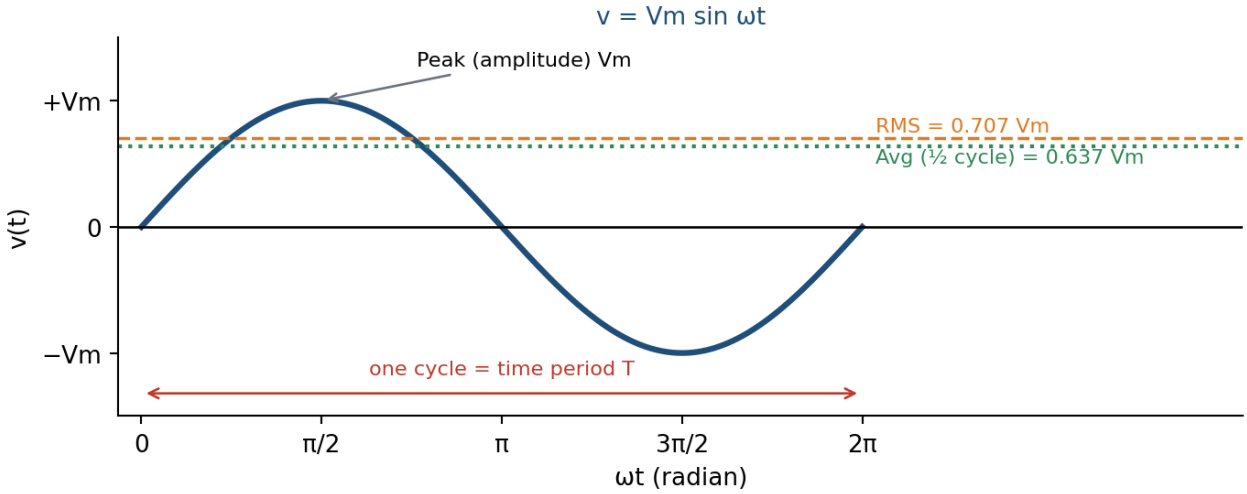


Fig 2.1 Sinusoidal waveform and its important values

$v = V_m \sin(\omega t)$ $i = I_m \sin(\omega t \pm \varphi)$ $\omega = 2\pi f = 2\pi/T$

Term	Meaning	Unit
Amplitude (peak)	Maximum value V_m or I_m	V or A
Time period T	Time for one complete cycle	s
Frequency f	Cycles per second, f = 1/T	Hz
Phase	Angular position of the wave at a given instant (ωt ± φ)	rad / degree
Phase difference φ	Angle by which one wave leads/lags another (Fig 2.2)	degree

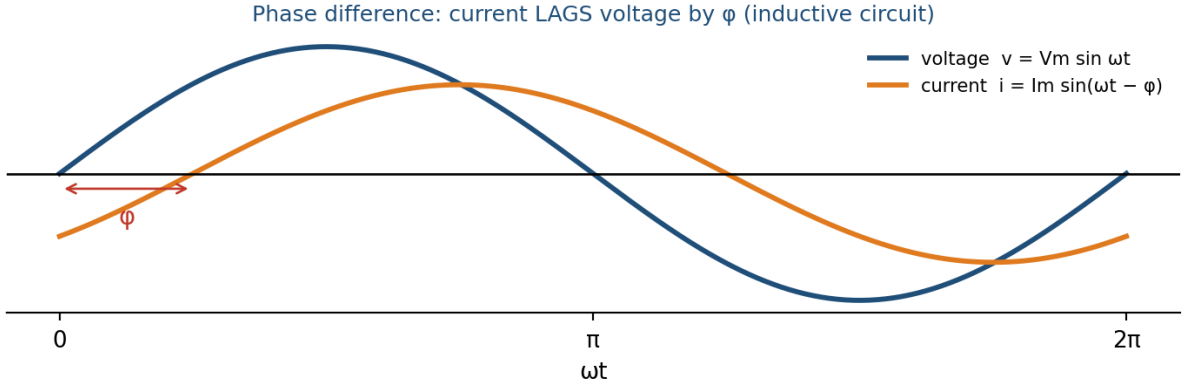


Fig 2.2 Phase difference between voltage and current

2.2 Average, RMS, Form Factor and Peak Factor

Quantity	Definition	Sine wave
Average value	Mean of instantaneous values over a *half* cycle (full-cycle average = 0)	$V_{av} = 2V_m/\pi =$ **0.637 V_m**
RMS (effective) value	Square root of mean of squares; the DC value giving the same heating effect	$V_{rms} = V_m/\sqrt{2} =$ **0.707 V_m**
Form factor	RMS value / Average value	**1.11**
Peak factor (crest)	Peak value / RMS value	**1.414**

SOLVED EXAMPLE

Q3. v = 311 sin 314t. Find V_{rms}, f and T.
$V_m = 311$ V ⇒ $V_{rms} = 0.707 \times 311 =$ **220 V**; ω = 314 ⇒ f = 314/2π = **50 Hz**; T = 1/50 = **20 ms**. Average (half-cycle) = 0.637 × 311 = 198 V.

2.3 AC through R, L and C (Phasor Diagrams)

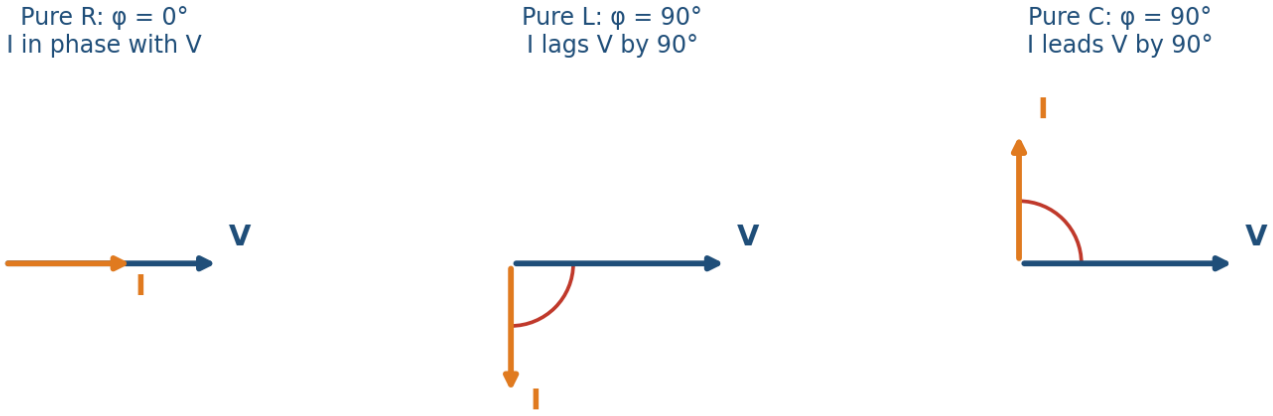


Fig 2.3 Phasor diagrams (voltage V is the reference along the x-axis)

Circuit	Opposition	Phase relation	Memory trick
Pure R	Z = R	I in phase with V (φ = 0°)	—
Pure L	$X_L = \omega L = 2\pi fL$	I **lags** V by 90°	"ELI": E before I in L
Pure C	$X_C = 1/\omega C = 1/2\pi fC$	I **leads** V by 90°	"ICE": I before E in C

Concept of Impedance (Z): the total opposition of a circuit to AC, combining resistance and reactance. For a series R-L-C circuit:

$$\mathbf{Z = \sqrt{[\ R^2 + (X_L - X_C)^2\ ]}\ \tan \varphi = (X_L - X_C) / R\ I = V / Z}$$

If $X_L > X_C$ the circuit is **inductive** (current lags); if $X_C > X_L$ it is **capacitive** (current leads); if $X_L = X_C$ it is **resonance** (Z = R, minimum).

SOLVED EXAMPLE

Q4. Find X_L of a 0.1 H coil and X_C of a 100 μF capacitor at 50 Hz.
$X_L = 2\pi \times 50 \times 0.1 =$ **31.4 Ω**; $X_C = 1/(2\pi \times 50 \times 100\times10^{-6}) =$ **31.8 Ω**. Note: X_L *increases* with frequency, X_C *decreases* with frequency (a capacitor passes high frequencies, an inductor blocks them).

APPLICATIONS

Series RLC resonance occurs at $f_0 = 1/(2\pi\sqrt{LC})$, where $X_L = X_C$, Z = R (minimum), current is maximum and pf = 1 — used in radio tuning.

2.4 Active, Reactive and Apparent Power; Power Factor

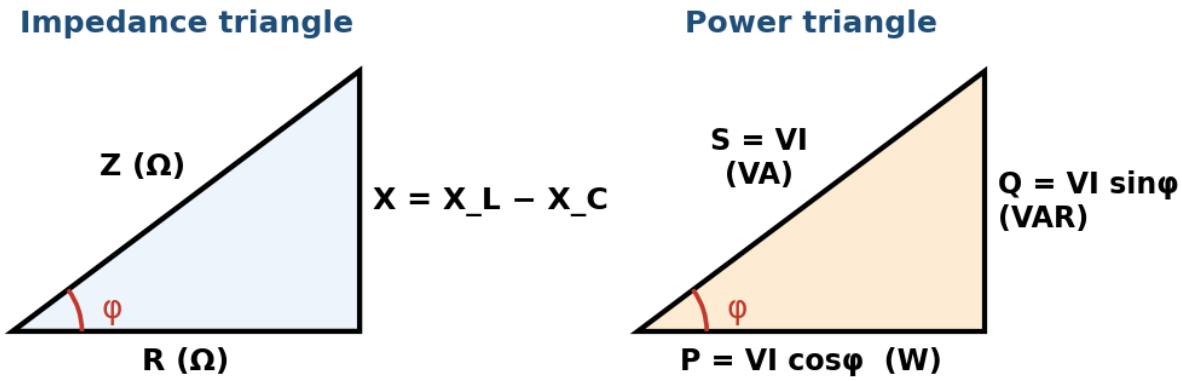


Fig 2.4 Impedance triangle and power triangle

Power	Symbol & formula	Unit	Meaning
Active (true)	P = VI cosφ = I²R	watt (W)	Real work done (heat, light, motion)
Reactive	Q = VI sinφ = I²X	VAR	Energy swapped between source and L/C; no work
Apparent	S = VI = I²Z	VA	Total power supplied; S² = P² + Q²

DEFINITION

Power factor = cos φ = R/Z = P/S (lies between 0 and 1; lagging for inductive loads, leading for capacitive loads). A **low power factor** means higher current for the same power → more losses and larger cables; it is improved by connecting **capacitors** across inductive loads.

SOLVED EXAMPLE

Q5. A series circuit has R = 30 Ω and X_L = 40 Ω across 230 V, 50 Hz. Find Z, I, pf, P, Q, S.
Z = √(30² + 40²) = **50 Ω**; I = 230/50 = **4.6 A**; φ = $\tan^{-1}(40/30)$ = 53.1°; pf = 30/50 = **0.6 lagging**.
P = 230 × 4.6 × 0.6 = **635 W** (check: I²R = 21.16 × 30 = 635 ✓); Q = 230 × 4.6 × 0.8 = **846 VAR**; S = 230 × 4.6 = **1058 VA**.

3 QUICK REVISION & EXAM CORNER

Topic	Key formulas
Ohm’s law	$V = IR$
Series / Parallel	$R_{eq} = \Sigma R$ / $1/R_{eq} = \Sigma(1/R)$
KCL / KVL	$\Sigma I = 0$ at node / $\Sigma V = 0$ in loop
AC wave	$\omega = 2\pi f$, $T = 1/f$, $V_{rms} = 0.707V_m$, $V_{av} = 0.637V_m$, FF = 1.11, PF(peak) = 1.414
Reactance & Z	$X_L = 2\pi fL$, $X_C = 1/2\pi fC$, $Z = \sqrt{(R^2 + X^2)}$
Power	$P = VI\cos\varphi$, $Q = VI\sin\varphi$, $S = VI$, $pf = \cos\varphi$

Short questions (2 marks)

- State Ohm’s law. Mention two limitations.
- State KCL and KVL.
- Define RMS value, form factor and peak factor of a sine wave.
- What is the phase relation between V and I in a pure inductor and a pure capacitor?
- Define impedance and power factor.
- Why is the RMS value used to rate AC supplies?

Long questions (5–10 marks)

- Explain superposition theorem with a neat example circuit.
- Derive RMS and average values of a sinusoidal current.
- Draw phasor diagrams for pure R, L and C circuits and explain.
- Explain active, reactive and apparent power with the power triangle.
- Three resistors 5 Ω, 10 Ω and 20 Ω are connected in (a) series and (b) parallel across 40 V. Find total current in each case. *(Ans: 1.14 A; 14 A)*

Practice numericals (with answers)

No.	Problem	Answer
1	Two resistors 6 Ω and 3 Ω are in parallel across 18 V. Find total current.	9 A (R_{eq} = 2 Ω)
2	Find RMS and average value of a sine wave with peak 200 V.	141.4 V; 127.4 V
3	R = 8 Ω in series with X_L = 6 Ω across 100 V. Find I and pf.	10 A; 0.8 lag
4	A load draws 5 A at 230 V with pf 0.8. Find P and Q.	920 W; 690 VAR
5	Find the series equivalent of 10 Ω, 15 Ω and 25 Ω, and current at 100 V.	50 Ω; 2 A

Quick MCQs

Question	Options	Ans
1. Unit of inductance is	(a) farad (b) henry (c) ohm (d) weber	b
2. Which law is based on conservation of charge?	(a) KVL (b) Ohm (c) KCL (d) Faraday	c
3. Form factor of a sine wave is	(a) 1.414 (b) 1.11 (c) 0.707 (d) 0.637	b
4. In a pure capacitor, current ... voltage by 90°	(a) lags (b) leads (c) is in phase with (d) is zero	b
5. Power factor of a pure inductor is	(a) 1 (b) 0.5 (c) 0 (d) infinity	c
6. Voltage source is replaced by ... in superposition	(a) open circuit (b) short circuit (c) resistor (d) capacitor	b

UNIT 2

Machines & Measuring Instruments

Topics: DC generator • DC motor • Single-phase transformer • Three-phase induction motor • Alternator • Applications of machines • PMMC instrument • Moving-iron instrument • Wheatstone bridge

START HERE

Big picture: An **electrical machine** converts energy. A **generator** turns mechanical energy into electrical; a **motor** does the reverse; a **transformer** changes voltage level with no moving parts. **Measuring instruments** let us read current, voltage and resistance.

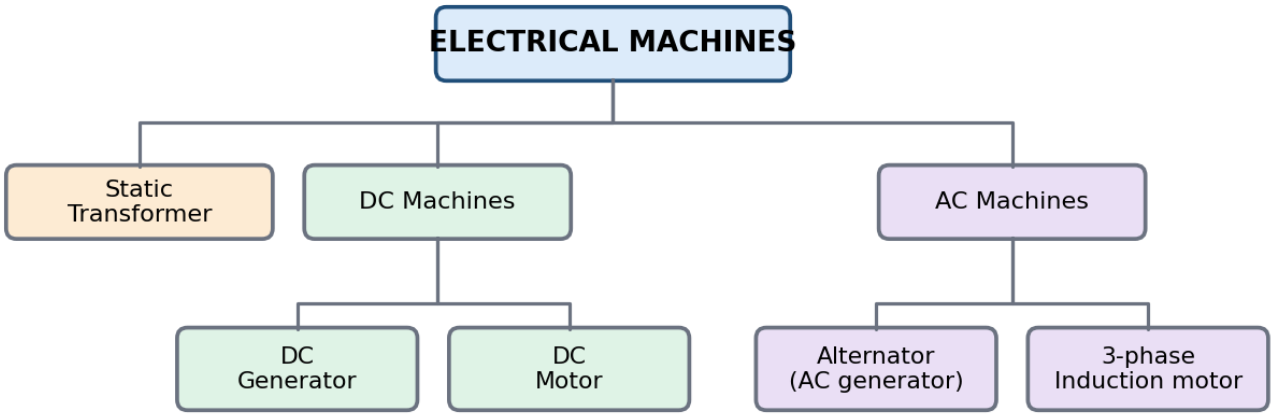


Fig 1 Classification of the machines in this unit

1 ELECTRICAL MACHINES

BASIC PRINCIPLES

Two golden principles

① **Faraday's law (generator action):** a conductor moving in a magnetic field (or a changing flux) induces an emf: e = B l v (or e = −N dΦ/dt).

② **Force on a current-carrying conductor (motor action):** F = B I l newton.

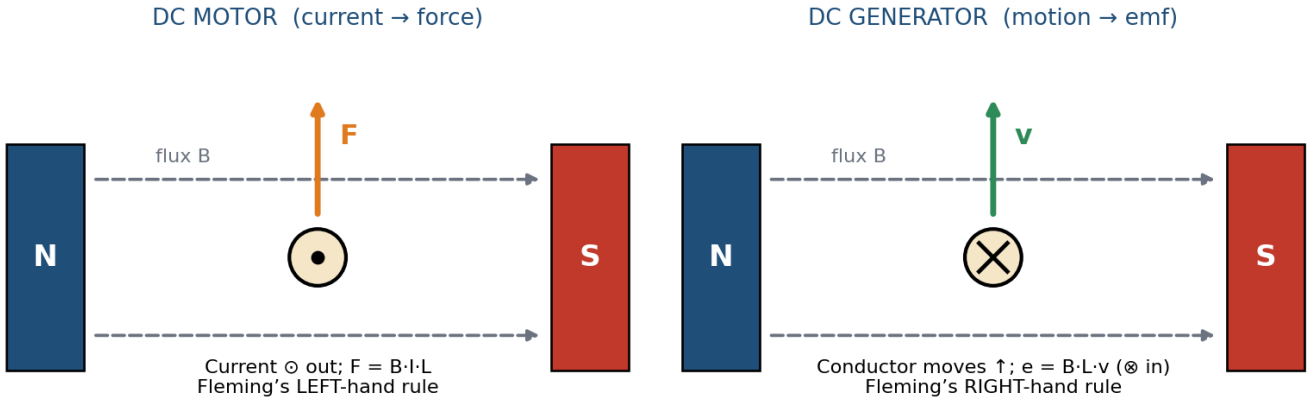


Fig 2 Motor action and generator action (same machine can do both)

1.1 DC Generator

Principle: converts mechanical energy into electrical energy (DC) by electromagnetic induction. The armature conductors cut the flux of the field poles and an emf is induced.

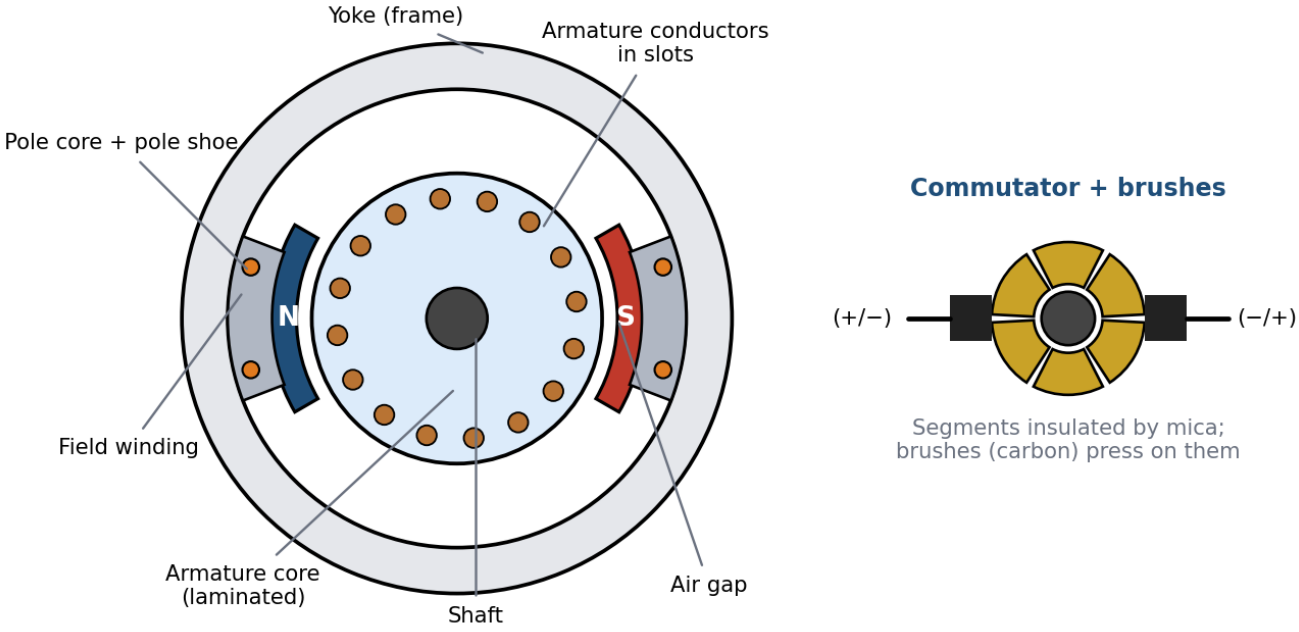


Fig 3 Cross-section of a DC machine and its commutator (same construction for generator and motor)

Part	Function
Yoke (frame)	Outer cast-iron/steel body; mechanical support and path for magnetic flux
Pole core + pole shoe	Carries field winding; shoe spreads flux evenly over the armature
Field winding	Copper coils on poles; produce the main magnetic field
Armature core + winding	Laminated steel core with slots; conductors in which emf is induced
Commutator	Copper segments; converts the alternating emf in the armature into **unidirectional** (DC) output
Brushes	Carbon blocks that collect current from the rotating commutator

$$E = (P \Phi Z N) / (60 A) \text{ volts}$$

P = number of poles, Φ = flux per pole (Wb), Z = total armature conductors, N = speed (rpm), A = number of parallel paths (A = P for lap winding, A = 2 for wave winding).

Type	Field connection	Typical applications
Separately excited	Field from an external DC source	Precise voltage control, test labs
Shunt	Field in **parallel** with armature	Battery charging, lighting, general supply
Series	Field in **series** with armature	Arc lamps, boosters
Compound	Both shunt and series fields	Lighting + power supply, welding sets

1.2 DC Motor

DEFINITION

Principle: when a current-carrying armature conductor is placed in a magnetic field it experiences a force; the force on the two sides of a coil forms a **torque** that rotates the armature. (Direction by Fleming's **left-hand rule**.)

$$T \propto \Phi I_a \quad E_b = V - I_a R_a \quad N \propto E_b / \Phi$$

Back emf (E_b): the rotating armature also generates an emf that *opposes* the supply voltage. It limits the armature current and makes the motor self-regulating — at start $E_b = 0$ so the current is very high, hence a **starter** is needed for large motors.

Motor	Characteristic	Applications
Shunt	Nearly **constant speed**	Lathes, fans, blowers, pumps, machine tools
Series	**High starting torque**; speed falls with load; never start on no-load (runaway)	Electric trains, cranes, hoists, elevators
Compound	Good starting torque + fairly steady speed	Rolling mills, presses, shears, punches

1.3 Single-Phase Transformer

DEFINITION
A **transformer** is a **static** device that transfers electrical energy from one circuit to another through **mutual induction**, changing voltage and current levels but keeping the **frequency** the same. It does **not** work on DC.

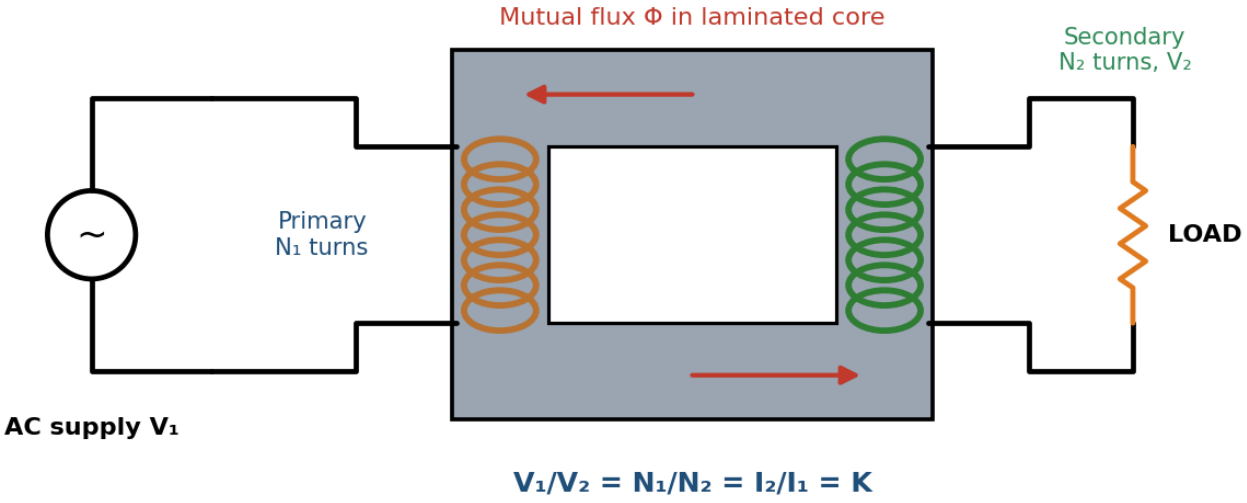


Fig 4 Single-phase transformer

Construction: a core of thin **laminated silicon-steel sheets** (reduces eddy-current loss) with two windings: **primary** (connected to supply) and **secondary** (connected to load). Types: *core type* and *shell type*. Small units are air-cooled; large ones are **oil-immersed**.
Working: AC in the primary produces an alternating flux Φ in the core → this flux links the secondary → emf is induced in the secondary by mutual induction.

$$E = 4.44\, f\, N\, \Phi_m \qquad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = K$$

Efficiency η = (Output power / Input power) × 100

Step-up transformer	Step-down transformer
$N_2 > N_1$, $K > 1$, $V_2 > V_1$, $I_2 < I_1$	$N_2 < N_1$, $K < 1$, $V_2 < V_1$, $I_2 > I_1$
Used at power stations (11 kV → 400 kV)	Used in distribution (11 kV → 415/230 V), adapters

REMEMBER
Losses: *Iron loss* (hysteresis + eddy current, constant) and *copper loss* (I^2R, varies with load). Ideal transformer: $V_1I_1 = V_2I_2$.

SOLVED EXAMPLE
Q1. A 2200/220 V transformer has 1000 primary turns. Find secondary turns and the primary current when it supplies 10 A.
$N_2 = N_1 \times V_2/V_1 = 1000 \times 220/2200 =$ **100 turns**. $I_1 = I_2 \times (V_2/V_1) = 10 \times 0.1 =$ **1 A**.

APPLICATIONS
Power transmission & distribution, mobile chargers, isolation, impedance matching, welding, voltage stabilisers.

1.4 Three-Phase Induction Motor

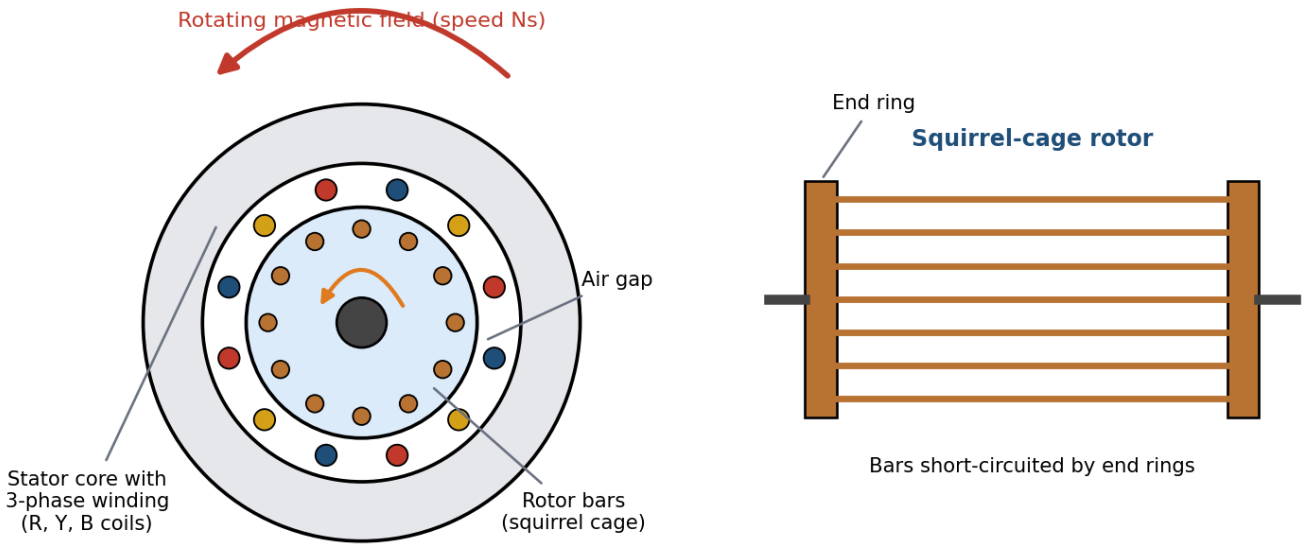


Fig 5 Stator and squirrel-cage rotor of a three-phase induction motor

DEFINITION

Principle: A three-phase supply to the stator winding creates a **rotating magnetic field** of synchronous speed N_s. It cuts the rotor conductors and induces emf → current → force (Lenz's law) → the rotor turns **in the direction of the field**, but always slightly slower. No electrical connection to the rotor, so it is also called an **asynchronous** motor.

$$N_s = 120\,f / P \text{ (rpm)} \quad \text{Slip } s = (N_s - N)/N_s$$

	Squirrel-cage rotor	Slip-ring (wound) rotor
Rotor	Copper/aluminium bars shorted by end rings	3-phase winding brought out to slip rings
Starting torque	Low	High (external resistance added)
Cost / maintenance	Cheap, rugged, no brushes	Costlier, needs brush maintenance
Uses	Pumps, fans, compressors, conveyors (≈90% of motors)	Cranes, lifts, crushers

SOLVED EXAMPLE

Q2. A 4-pole, 50 Hz induction motor runs at 1440 rpm. Find slip.

$N_s = 120 \times 50/4 = 1500$ rpm; $s = (1500 - 1440)/1500 = 0.04 =$ **4 %**.

1.5 Alternator (Synchronous Generator)

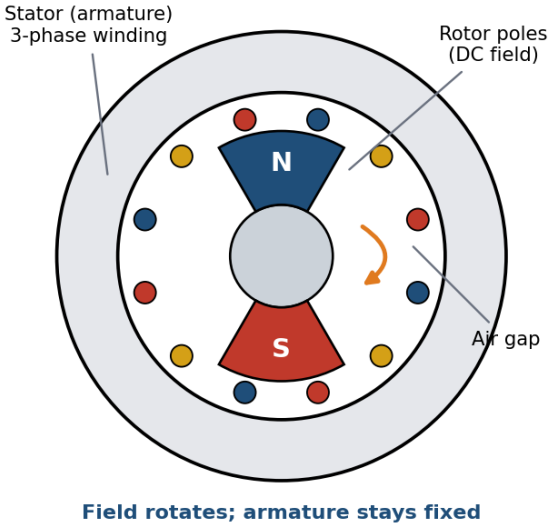


Fig 6a Cross-section of an alternator

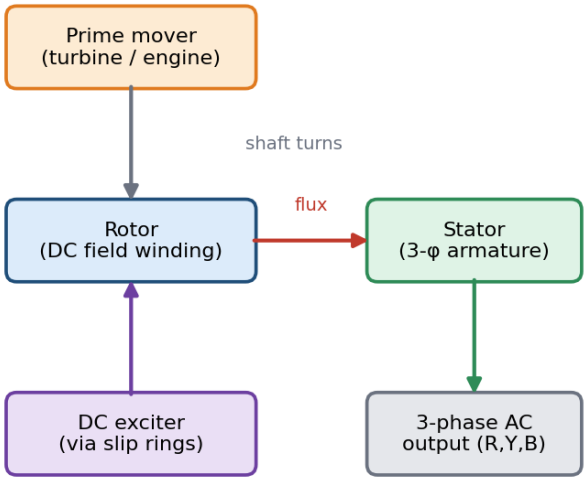


Fig 6b Block diagram of working

DEFINITION

Principle: Electromagnetic induction. The **rotor** carries a DC-excited field winding (supplied through slip rings); when turned by a prime mover its flux cuts the **stationary armature (stator)** conductors and induces a 3-phase AC emf.

$$f = P\,N / 120\ (Hz)\quad E_{ph} = 4.44\,K\,f\,\Phi\,T_{ph}$$

- **Why stationary armature?** Heavy high-voltage output (11 kV+) is easier to insulate and tap without slip rings; slip rings carry only the small DC field current.
- **Rotor types:** *Salient-pole* (low speed, hydro plants, many poles) and *cylindrical/non-salient* (high speed, steam/gas turbines, 2–4 poles).
- **Synchronous speed:** the rotor must run at N = 120f/P to deliver the correct frequency (50 Hz).

1.6 APPLICATIONS OF ELECTRICAL MACHINES

Machine	Typical applications
DC generator	Battery charging, electroplating, excitation of alternators, welding
DC motor	Electric traction (trains, trams, EVs), cranes, elevators, rolling mills, printing presses
Transformer	Power grid, adapters, inverters, isolation, instrument transformers
3-phase induction motor	Industrial drives, water pumps, fans, compressors, lifts, conveyors, washing machines
Alternator	Thermal, hydro and nuclear power stations, diesel generator sets, wind turbines, vehicle charging (alternator)

2 MEASURING INSTRUMENTS

Instruments are classified as **indicating** (show value on a scale), **recording** and **integrating** (energy meters). Every deflection instrument needs three torques: **deflecting**, **controlling** (spring) and **damping** (to stop oscillation).

2.1 Permanent Magnet Moving Coil (PMMC) Instrument

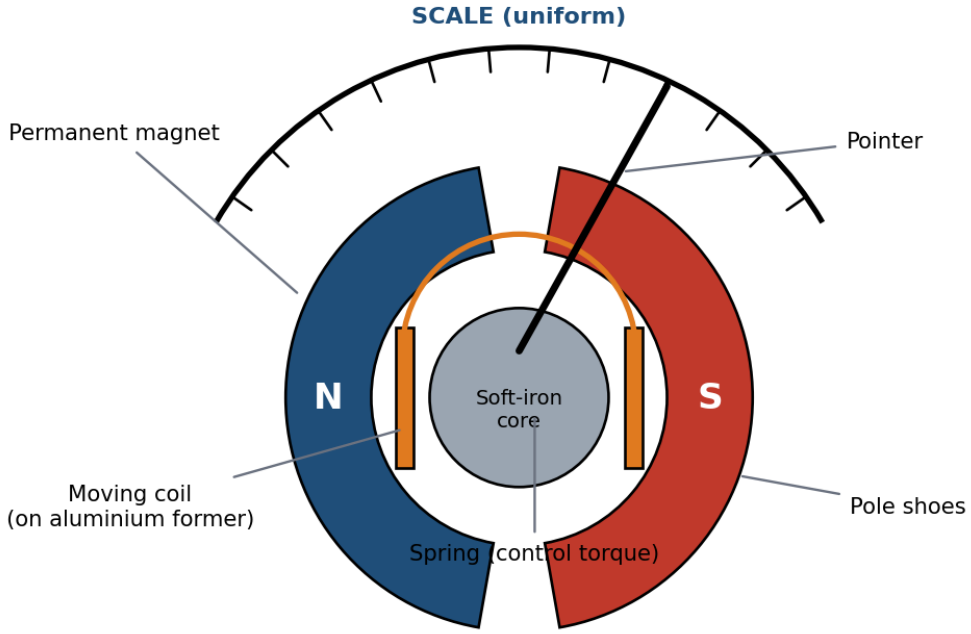


Fig 7 PMMC instrument

DEFINITION
Principle: a current-carrying coil placed in the magnetic field of a permanent magnet experiences a torque ∝ current. Deflection θ ∝ I, so the scale is **uniform**. Controlling torque: spiral **springs**. Damping: **eddy currents** in the aluminium former.

- **Advantages:** very accurate, uniform scale, high sensitivity, low power consumption, not affected by stray fields.
- **Disadvantages:** works on **DC only** (on AC the torque reverses and the pointer just vibrates), costlier, delicate.
- **Use:** DC ammeter (with parallel **shunt**) and DC voltmeter (with series **multiplier** resistor).

2.2 Moving Iron (MI) Instrument

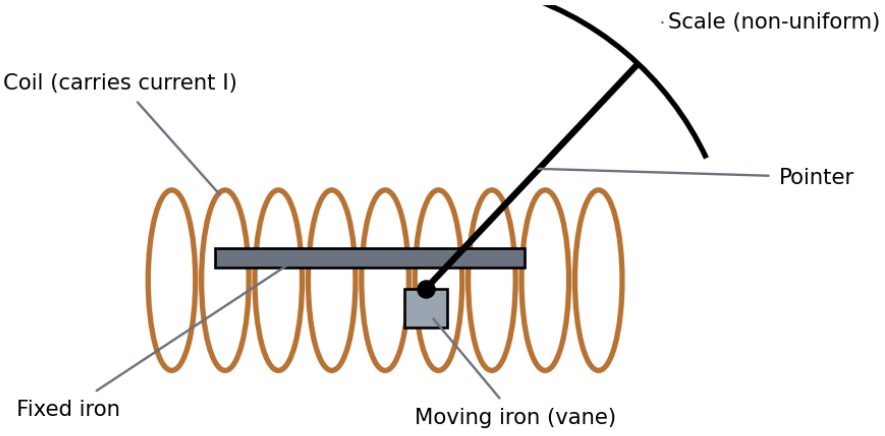


Fig 8 Moving-iron (attraction type) instrument

DEFINITION
Principle: a soft-iron piece placed in the field of a current-carrying coil gets magnetised and is **attracted** (attraction type) or two similar irons **repel** (repulsion type). Deflecting torque $\propto I^2$, so the scale is **non-uniform** (crowded at the start) but the instrument works on **both AC and DC**. Damping: air friction.

Feature	PMMC	Moving iron
Works on	DC only	AC and DC
Deflection $\propto$	I (uniform scale)	I^2 (non-uniform scale)
Accuracy / cost	High / costly	Moderate / cheap, rugged
Power consumption	Very low	Higher
Damping	Eddy current	Air friction

2.3 Wheatstone Bridge

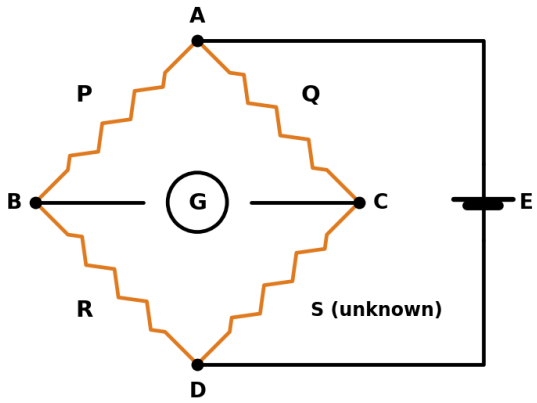


Fig 9 Wheatstone bridge

Use: measuring an unknown medium resistance accurately by comparison with known resistances.

Balance: adjust R until the galvanometer shows **zero** deflection. Then $V_B = V_C$ and the ratio arms satisfy the condition below.

P, Q = ratio arms; R = variable known resistance; S = unknown resistance.

$$P / Q = R / S \Rightarrow S = R \times Q / P$$

SOLVED EXAMPLE
Q3. At balance P = 100 Ω, Q = 1000 Ω, R = 35 Ω. Then S = 35 × 1000/100 = **350 Ω**.

- **Applications:** precise resistance measurement, strain gauges, temperature (RTD) and sensor circuits.
- **Limitations:** unsuitable for very low or very high resistance; heating by current may alter resistance.

3 QUICK REVISION & EXAM CORNER

Topic	Remember
DC generator emf	E = PΦZN/60A; commutator gives DC
DC motor	Eb = V − IaRa; T ∝ ΦIa; series = high starting torque
Transformer	$V_1/V_2 = N_1/N_2 = I_2/I_1$; E = 4.44fNΦm; no DC, no frequency change
Induction motor	Ns = 120f/P; slip = (Ns − N)/Ns; rotor never reaches Ns
Alternator	f = PN/120; rotating field, stationary armature
PMMC vs MI	PMMC: DC, uniform scale; MI: AC/DC, I^2 scale
Wheatstone	P/Q = R/S at null deflection

Short questions (2 marks)

- State the working principle of a DC motor and name Fleming's rule used.
- What is the function of a commutator?
- What is back emf? Why is a starter needed for a DC motor?
- Why is the transformer core laminated?
- Define slip of an induction motor.
- Why do alternators have a stationary armature?
- Give two advantages of PMMC instruments.
- Write the balance condition of a Wheatstone bridge.

Long questions (5–10 marks)

- Explain construction and working of a DC generator with a neat diagram; derive its emf equation.
- Explain the construction and working principle of a single-phase transformer; derive E = 4.44 f N Φm.
- Explain the production of a rotating magnetic field and working of a three-phase induction motor.
- Describe the construction and working of an alternator.
- With neat diagrams compare PMMC and moving-iron instruments.
- Explain the Wheatstone bridge and derive its balance condition.

Practice numericals (with answers)

No.	Problem	Answer
1	A 6-pole DC generator runs at 1000 rpm, Φ = 0.02 Wb, Z = 300, lap winding. Find emf.	E = 6×0.02×300×1000/(60×6) = 100 V
2	A transformer has N_1 = 500, N_2 = 50, V_1 = 440 V. Find V_2.	44 V
3	A 6-pole, 50 Hz induction motor has slip 5 %. Find rotor speed.	N_s = 1000 rpm; N = 950 rpm
4	Alternator with 4 poles must give 50 Hz. Find speed.	1500 rpm
5	DC motor: V = 230 V, I_a = 20 A, R_a = 0.5 Ω. Find E_b.	220 V

Quick MCQs

Question	Options	Ans
1. A transformer works on	(a) DC (b) AC (c) both (d) neither	b
2. Induction motor runs at … synchronous speed	(a) above (b) equal to (c) below (d) double	c
3. Speed of a series motor on no-load is	(a) zero (b) constant (c) dangerously high (d) low	c
4. Scale of an MI instrument is	(a) uniform (b) non-uniform (c) logarithmic (d) none	b

UNIT 3

Energy Resources, Electricity Bill & Safety Measures

Topics: Conventional & non-conventional resources • Hydel, nuclear, solar & wind generation • Appliance power ratings • Unit (kWh) • Two-part tariff • Bill calculation • Fuse & MCB • Electric shock • Earthing • Safety precautions

START HERE
Big picture: Electricity must be **generated** from some energy resource, **paid for** by consumers (the electricity bill) and **used safely**. This unit covers all three.

1 ENERGY RESOURCES

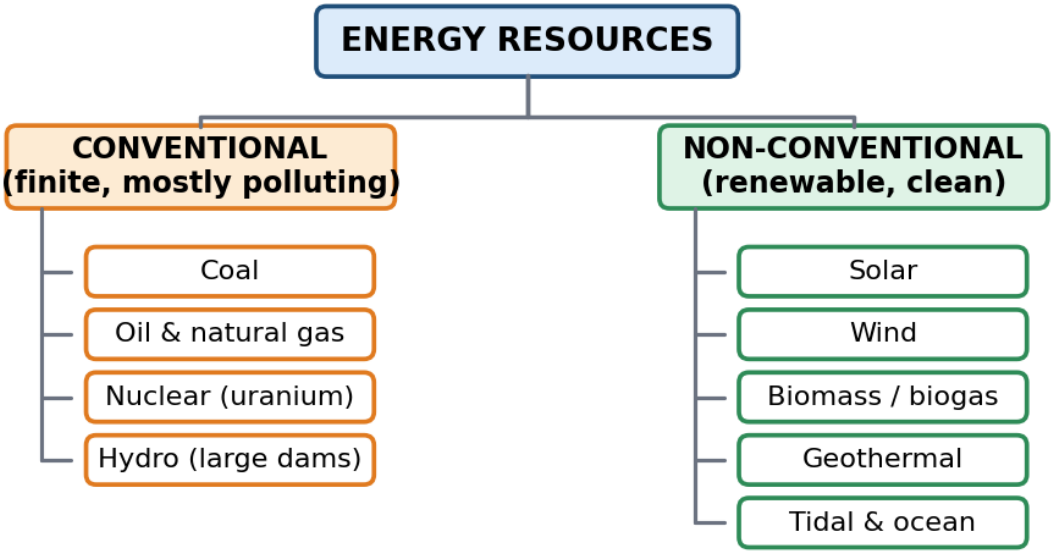


Fig 1 Classification of energy resources

Feature	Conventional	Non-conventional
Availability	Limited (fossil fuels, uranium); being depleted	Unlimited; renewable
Pollution	High (CO_2, SO_2, ash, nuclear waste)	Very low / nil
Cost	High running cost, low set-up cost (thermal)	High set-up cost, almost free fuel
Power level	Large, steady, reliable	Smaller, depends on weather/location
Examples	Coal, oil, gas, nuclear, large hydro	Solar, wind, biomass, geothermal, tidal

1.1 Hydel (Hydroelectric) Power Generation

DEFINITION

Working idea: potential energy of water stored at a height → kinetic energy as it falls through the **penstock** → mechanical energy in the **turbine** → electrical energy in the **generator**.

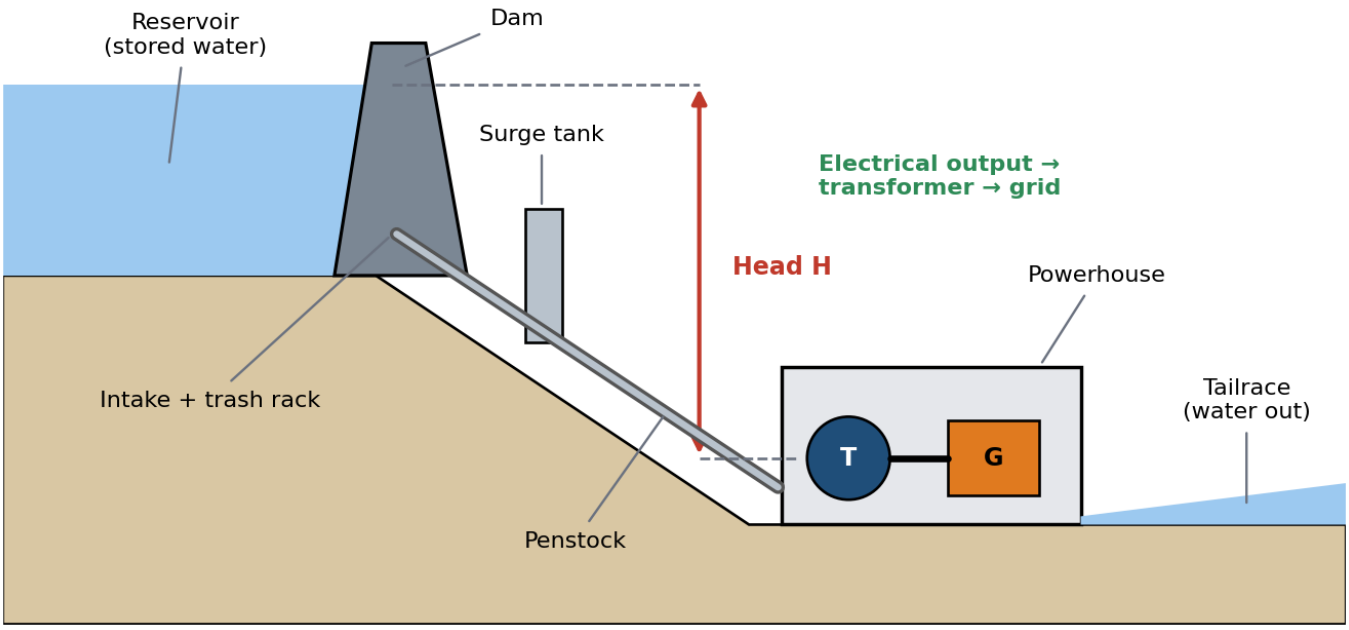


Fig 2 Layout of a hydroelectric power plant

$P = 9.81 \times Q \times H \times \eta$ kW
(Q = flow in m³/s, H = head in m, η = overall efficiency)

Component	Function
Dam + reservoir	Stores water and creates the head H
Intake / trash rack	Admits water and filters out debris
Penstock	Steel pipe carrying water to the turbine
Surge tank	Absorbs pressure surges (water hammer) when load changes
Turbine	**Pelton** (high head), **Francis** (medium head), **Kaplan** (low head)
Tailrace	Channel returning used water to the river

- **Merits:** no fuel cost, clean, long life, quick start, also gives irrigation and flood control.
- **Demerits:** very high initial cost, depends on rainfall, needs large land area, displaces people/ecology.

1.2 Nuclear Power Generation

DEFINITION

Working idea: In the reactor, neutrons split enriched **uranium-235** atoms (**nuclear fission**); each fission releases heat and more neutrons, sustaining a **chain reaction**. The heat makes steam, which drives a turbine-generator, just like a thermal plant but with the reactor replacing the boiler.

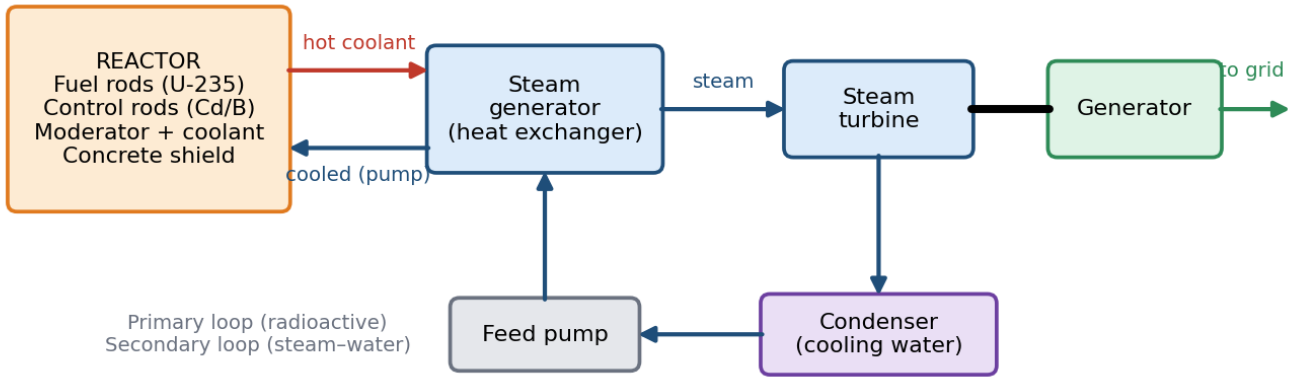


Fig 3 Block diagram of a nuclear power plant

Part	Role
Fuel rods	Enriched uranium (U-235) pellets; source of fission heat
Moderator	Slows neutrons so they cause fission (heavy water, graphite, ordinary water)
Control rods	Cadmium / boron absorb neutrons; inserted to slow or stop the reaction
Coolant	Water / heavy water / CO_2 carries heat to the steam generator
Shielding	Thick concrete + steel to stop radiation

- **Merits:** huge energy from a tiny amount of fuel, no CO_2 during operation, steady base-load power.
- **Demerits:** radioactive waste disposal, very high capital cost, risk of accidents, strict safety needs.

1.3 Solar Power Generation

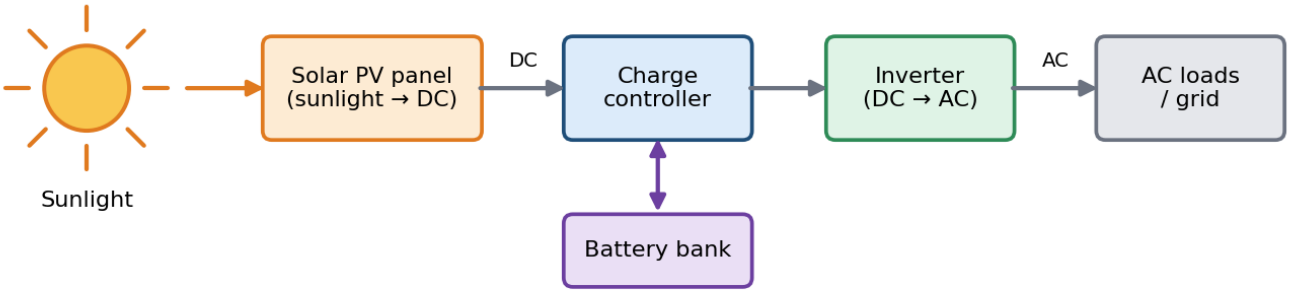


Fig 4 Stand-alone solar photovoltaic (PV) system

A **solar cell** (silicon p-n junction) converts sunlight directly into **DC** by the **photovoltaic effect**. Many cells form a module; modules form an array. The **charge controller** protects the battery, the **battery** stores energy for night use and the **inverter** converts DC to AC. In *grid-connected* systems the battery is omitted and extra power is fed to the grid.

- **Merits:** free, clean, silent, low maintenance, good for remote areas.

- **Demerits:** works only in daylight, high initial cost, needs large area, output varies with weather.

1.4 Wind Power Generation

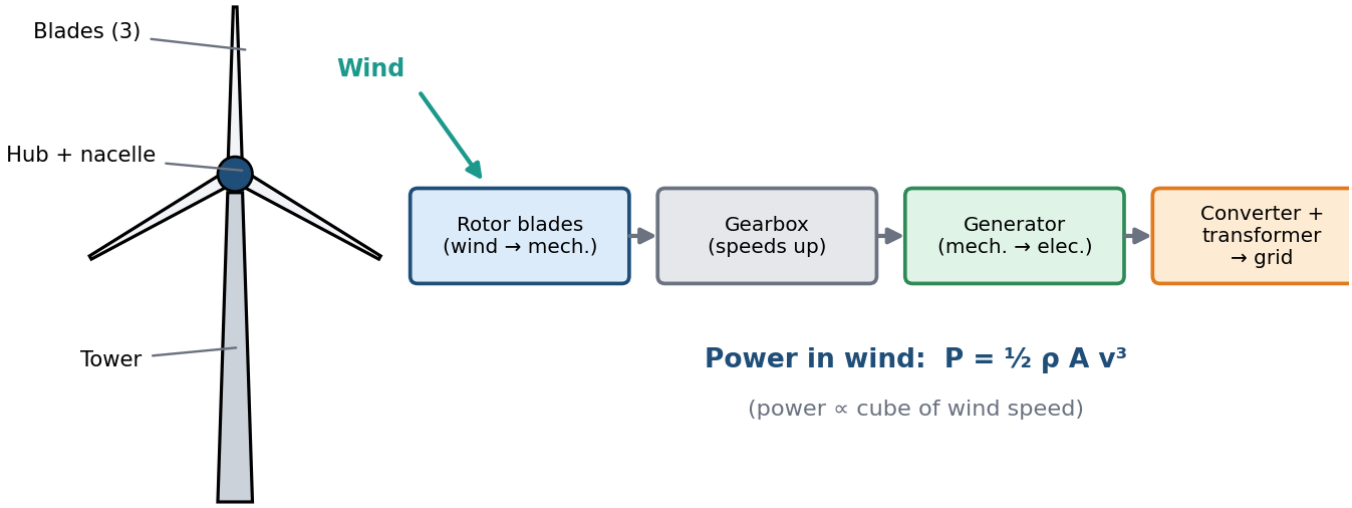


Fig 5 Wind turbine and energy-conversion chain

Moving air turns the turbine blades; the slow rotor speed is increased by a **gearbox** to drive the **generator**. A yaw system turns the nacelle to face the wind. Typical cut-in speed ≈ 3–4 m/s; best sites have steady wind > 6 m/s.

- **Merits:** free and clean fuel, land below can still be farmed, quick to install.
- **Demerits:** wind is intermittent, noise, bird hazard, needs windy sites, high initial cost.

2 ELECTRICITY BILL

2.1 Power Rating of Household Appliances

The **power rating** (in watt, W) is printed on every appliance. Typical (approximate) values:

Appliance	Typical power	Appliance	Typical power
LED bulb	5 – 12 W	Refrigerator	150 – 250 W
Tube light (LED)	20 – 40 W	Washing machine	400 – 600 W
Ceiling fan	60 – 80 W	Electric iron	750 – 1000 W
LED TV	60 – 150 W	Water heater (geyser)	1500 – 3000 W
Desktop PC (with monitor)	150 – 300 W	Microwave oven	1000 – 1400 W
Laptop	40 – 65 W	Air conditioner 1.5 ton	1200 – 1800 W
Printer (laser, printing)	300 – 500 W	Wi-Fi router	5 – 15 W

2.2 The “Unit” of Electrical Energy

DEFINITION
1 unit = 1 kilowatt-hour (kWh) = the energy used by a 1000 W (1 kW) load running for 1 hour. (1 kWh = 3.6×10^6 joule.)

Energy (kWh) = Power (W) × Time (h) / 1000



Fig 6 Steps to calculate an electricity bill

2.3 Two-Part Tariff

The consumer pays two parts: **(1) Fixed (demand) charge** – depends on connected load/sanctioned load in kW, charged per month whether or not energy is used; **(2) Energy charge** – depends on units (kWh) consumed, charged at a rate per unit (often in slabs).

Bill = Fixed charge + (Units consumed × Rate per unit) [+ taxes/duty]

SOLVED EXAMPLE
Q1. Monthly (30-day) bill for a house with: 6 LED lamps (10 W, 5 h/day), 3 fans (75 W, 8 h/day), 1 TV (100 W, 4 h/day), 1 refrigerator (200 W, runs 8 h/day), 1 laptop (60 W, 5 h/day). *Assume* fixed charge ₹100/month, energy charge ₹6/unit.
Lamps: 6×10×5/1000 = 0.30 kWh | Fans: 3×75×8/1000 = 1.80 | TV: 100×4/1000 = 0.40 | Fridge: 200×8/1000 = 1.60 | Laptop: 60×5/1000 = 0.30
Daily = **4.40 kWh** → monthly = 4.4 × 30 = **132 units** → energy charge = 132 × 6 = ₹792 → **Bill = 792 + 100 = ₹892**.
Q2. A 1.5 kW air conditioner runs 8 h/day for 30 days: 1.5 × 8 × 30 = **360 units** (≈ ₹2160 at ₹6/unit).

NOTE
Tariff rates differ from state to state and change often — use the rates given in the question or on your own bill. Saving tip: use 5-star appliances and LED lamps, switch off standby devices.

3 EQUIPMENT SAFETY MEASURES

3.1 Fuse

DEFINITION
A **fuse** is a protective device with a thin wire (element) of low-melting alloy (tin-lead, zinc, silver, copper) connected **in series**. On excessive current, the **heating effect (I^2R)** melts the element and breaks the circuit.

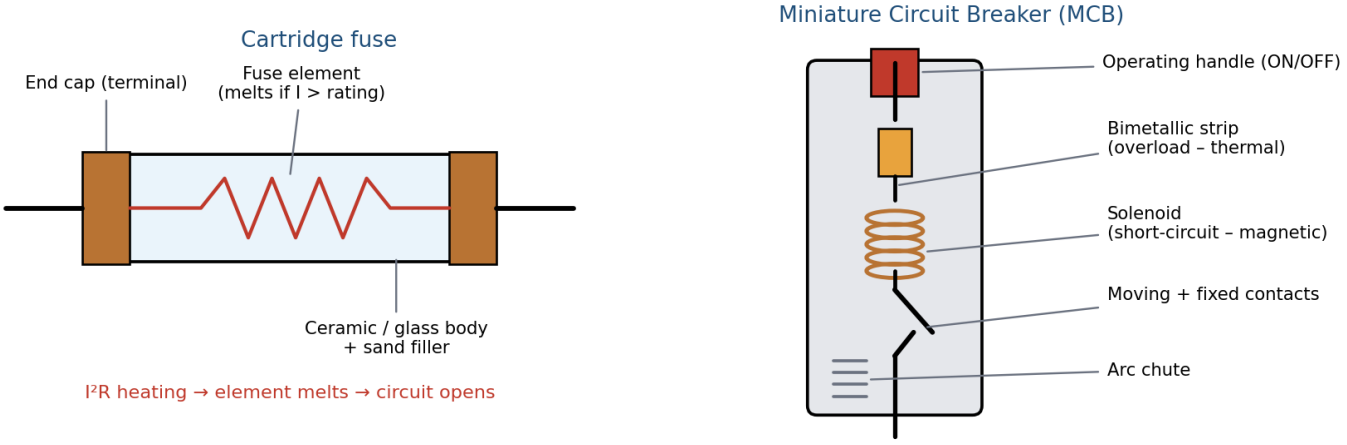


Fig 7 Cartridge fuse (left) and Miniature Circuit Breaker (right)

- **Types:** rewirable (kit-kat) fuse, cartridge (HRC) fuse.
- **Merits:** cheap, simple, no maintenance, very fast on heavy short circuits.
- **Demerits:** must be **replaced** after blowing, slow for small overloads, wrong wire rating can be dangerous, cannot be reset or remotely operated.

3.2 Miniature Circuit Breaker (MCB)

DEFINITION
An **MCB** is an automatic switch that **trips** on overload or short circuit and can be **reset** after the fault is cleared. **Overload:** heated **bimetallic strip** bends and releases the latch (delayed). **Short circuit:** the large current energises a **solenoid** whose plunger trips the contacts instantly. The arc formed is quenched in the **arc chute**.

Type	Trips at (× rated current)	Used for
B	3 – 5 ×	Lighting, resistive loads (homes)
C	5 – 10 ×	Small motors, ACs, mixed loads
D	10 – 20 ×	Transformers, large motors

	Fuse	MCB
Operation	Melts – one-time	Trips – resettable
Response	Slower on small overload	Fast, thermal + magnetic
Cost	Low	Higher
Fault indication	Need to inspect element	Handle position shows
Re-use	Replace element	Switch ON again

3.3 Personal Safety: Electric Shock

DEFINITION

Electric shock occurs when a part of the body completes a path between a live conductor and earth (or between two conductors) so that current flows through it. The **current** (not voltage alone) decides the damage: $I = V / R_{body}$.

Current through body (approx.)	Effect
≈ 1 mA	Threshold of feeling (tingling)
≈ 10 mA	Painful, muscles contract – may be unable to let go
≈ 30 mA	Breathing difficulty
50 – 100 mA	Ventricular fibrillation – may be fatal
> 200 mA	Severe burns, heart stops

Severity depends on: magnitude and duration of current, path through the body (hand-to-hand through chest is worst), skin condition (wet skin resistance ≈ 1 kΩ vs dry ≈ 100 kΩ) and supply type.

3.4 Earthing

DEFINITION

Earthing is the connection of the **non-current-carrying metal parts** of equipment (body, frame) to the general mass of earth through a low-resistance conductor. If insulation fails, the fault current flows safely to earth and the fuse/MCB trips — the body never stays at dangerous voltage.

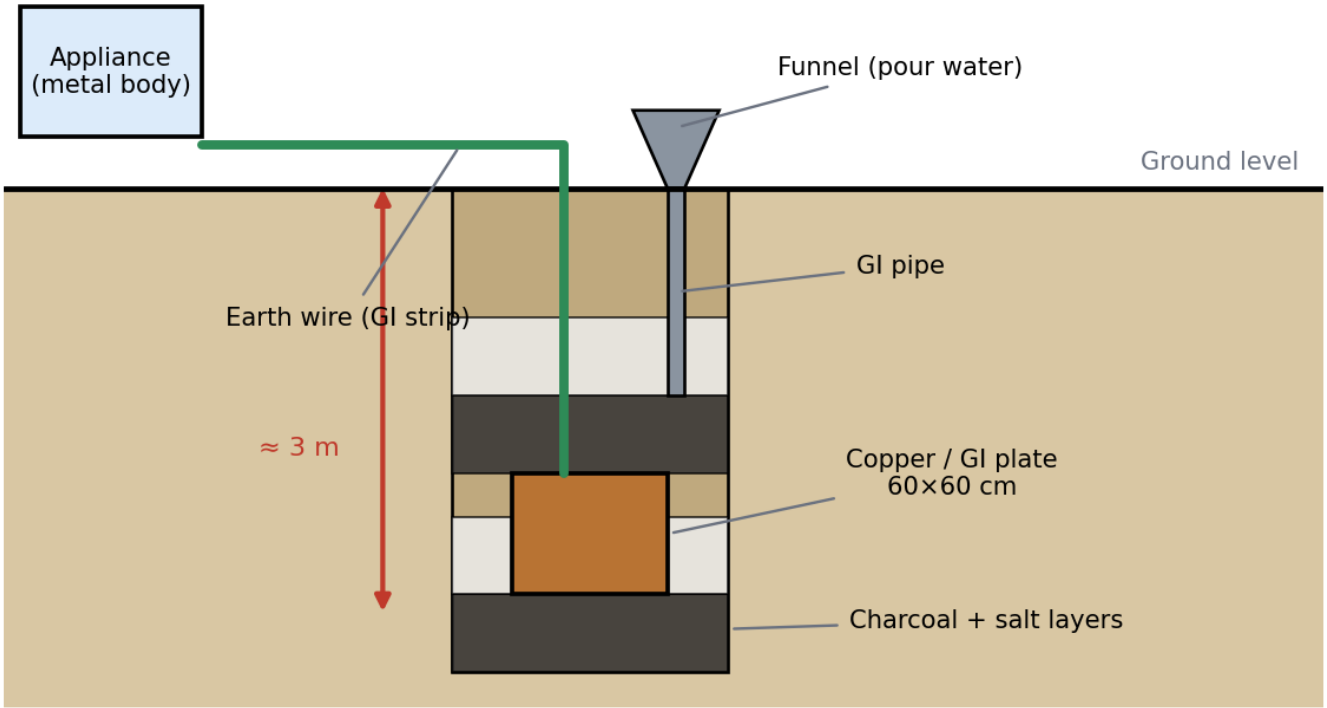


Fig 8 Plate earthing

Type	Details	Typical use
Plate earthing	Copper (60×60×3 mm) or GI (60×60×6 mm) plate buried ≈ 3 m deep with alternate layers of charcoal and salt	Substations, large buildings
Pipe earthing	GI pipe (≈ 38 mm dia, ≈ 2.5 m long) driven vertically; water poured through funnel to keep soil moist	Houses, small installations

Type	Details	Typical use
Rod earthing	Copper/GI rod (12–16 mm dia) driven into ground	Residential, telecom
Strip / wire earthing	Strip buried in trench ≈ 0.5 m deep	Rocky soil, overhead lines

REMEMBER
Earth resistance should be very low (typically < 5 Ω for homes). **Wire colours (India):** Live – Red/Brown, Neutral – Black/Blue, Earth – Green (or Green-Yellow). The earth pin of a 3-pin plug is longer and thicker.

3.5 Safety Precautions to Avoid Shock

- Always **switch off** the main supply before repairing; use a tester to confirm the line is dead.
- Use **proper earthing** and fit **MCB / ELCB (RCCB)** in every installation.
- Never touch switches or appliances with **wet hands** or while standing on a wet floor; use rubber-soled shoes / insulating mats.
- Use **ISI-marked** wires, plugs and appliances; replace damaged or bare wires immediately.
- Never use a fuse wire of the wrong rating; do not bridge a fuse with copper wire.
- Keep water away from sockets; use covered sockets in homes with children; do not overload sockets.
- If someone is stuck to a live wire: **switch off first** or push them away with a dry wooden stick – never touch them directly; then give first aid and call for medical help.

4 QUICK REVISION & EXAM CORNER

Topic	Remember
Hydel	P = 9.81 Q H η kW; Pelton – high head, Kaplan – low head
Nuclear	U-235 fission; moderator slows neutrons; control rods absorb neutrons
Solar / Wind	PV → DC → inverter; wind power ∝ v^3
Unit	1 kWh = 1000 W × 1 h; kWh = W × h / 1000
Bill	Fixed charge + units × rate
Fuse vs MCB	Fuse: I^2R melts, replace; MCB: bimetal + solenoid, resettable
Earthing	Low-resistance path for fault current; plate, pipe, rod, strip

Short questions (2 marks)

- Differentiate conventional and non-conventional energy sources.
- What is the function of the penstock and surge tank?
- What is the role of the moderator and control rods?
- Define 1 unit of electrical energy.
- What is a two-part tariff?
- Write any two demerits of a fuse.
- Why is earthing necessary?

Long questions (5–10 marks)

- Draw the layout of a hydroelectric power plant and explain its working.
- Explain the working of a nuclear power plant with a block diagram.
- Explain solar PV and wind power generation with neat diagrams.

- Explain the working of a fuse and MCB; compare them.
- What is earthing? Explain plate earthing with a neat sketch.
- Calculate the monthly bill for a given load list using a two-part tariff.

Practice numericals (with answers)

No.	Problem	Answer
1	A 2 kW heater runs 3 h daily. Find units used in 30 days.	2×3×30 = 180 units
2	A 40 W tube light and 75 W fan run 10 h daily. Units in 30 days?	(40+75)×10×30/1000 = 34.5 units
3	Fixed charge ₹120, rate ₹5/unit, consumption 200 units. Bill?	120 + 1000 = ₹1120
4	Hydel plant: Q = 20 m^3/s, H = 50 m, η = 80 %. Output power?	9.81×20×50×0.8 = 7848 kW ≈ 7.85 MW

BASIC ELECTRONICS ENGINEERING • PART B

UNIT I

Semiconductor Devices

From vacuum tubes to nano-electronics, diodes, Zener diode, BJT and the CE amplifier

WHAT YOU WILL LEARN

1 Evolution of electronics
2 PN junction diode & its characteristics
3 Zener effect & Zener diode
4 Bipolar Junction Transistor (BJT)
5 CB, CE, CC configurations
6 Input / output characteristics
7 Small-signal CE amplifier
8 Formulas, summary & exam questions

1 Evolution of Electronics: Vacuum Tubes to Nano-electronics

Electronics is the branch of engineering that controls the flow of electrons through vacuum, gases and (mostly today) **semiconductors** to process information and power. Every phone, TV and computer is a descendant of one idea: *use a small signal to control a large flow of electrons*.

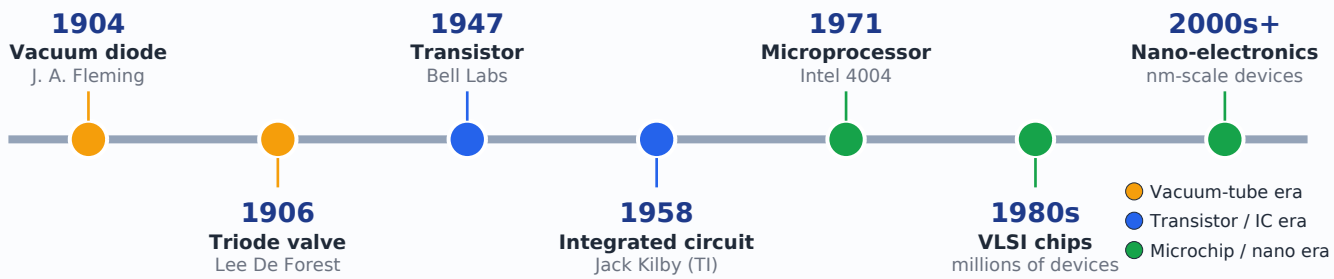


Fig 1.1 Milestones in the evolution of electronics

Vacuum tube: the first electronic device

- A heated **cathode** releases electrons (*thermionic emission*); a positive **anode (plate)** collects them. A **grid** in a triode controls the current, hence it amplifies.
- Used in early radio, TV, radar and the first computers (ENIAC, ~18,000 tubes).

Property	Vacuum tube	Transistor / IC
Size	Large and bulky	Tiny (nm to mm)
Power	High (heater needed)	Very low
Warm-up time	Needs seconds to heat	Instant on
Life and strength	Short life, fragile glass	Very long, rugged
Operating voltage	Hundreds of volts	A few volts
Cost (mass production)	High	Extremely low

Integration levels and nano-electronics

Era	Approx. transistors per chip	Example
SSI / MSI	up to a few hundred	Logic gates, counters
LSI	thousands to ~100,000	Early microprocessors, memories
VLSI / ULSI	lakhs to billions	Modern CPUs, mobile SoCs

- **Nano-electronics** deals with devices whose key dimensions are in the nanometre range (1–100 nm), where **quantum effects** become important.

- Examples: **FinFET / gate-all-around** transistors, **carbon-nanotube** and **graphene** devices, quantum dots, nanowires, memristors.
- Benefits: higher speed, lower power, huge density. Challenges: leakage, heat, tunnelling and fabrication cost.

Applications of nano-electronics
Smartphone processors, AI accelerators, flash memory, medical biosensors, flexible displays, solar cells.

Did you know?
- The ENIAC computer (1946) used about 18,000 vacuum tubes and filled a large hall; a modern phone chip packs **billions** of transistors in a fingernail-sized die.
- A human hair is ~80,000 nm wide, while today's smallest transistor features are only a few nanometres.

2 PN Junction Diode

A **semiconductor** (Si, Ge) conducts better than an insulator but worse than a conductor. Adding impurities (**doping**) makes it *extrinsic*.

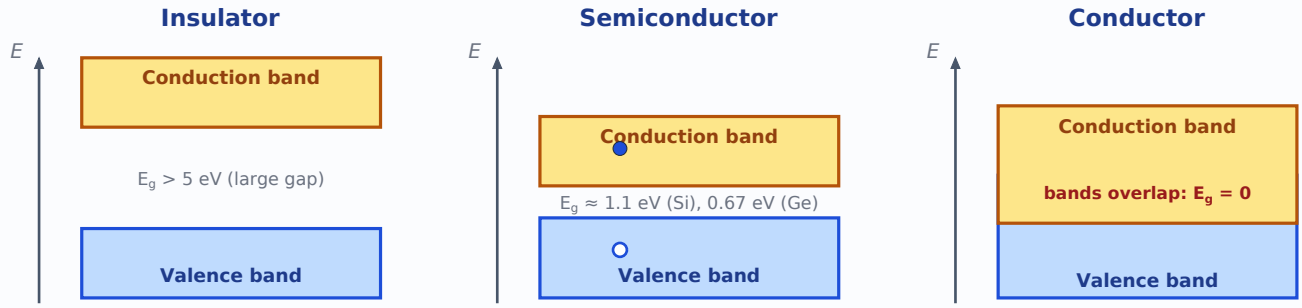


Fig 1.2 Energy-band view: the gap between valence and conduction bands decides how well a material conducts

	N-type	P-type
Dopant	Pentavalent (P, As, Sb): *donor*	Trivalent (B, Al, Ga): *acceptor*
Majority carriers	Free electrons	Holes
Minority carriers	Holes	Electrons
Charge of crystal	Neutral overall	Neutral overall

DEFINITION
A **PN junction diode** is formed by joining P-type and N-type semiconductors. It is a **two-terminal, one-way device**: Anode (P) and Cathode (N). Symbol: ▶| (current flows easily from anode to cathode).

Formation of the depletion region

- Electrons from N-side diffuse across to the P-side; holes diffuse the other way and they **recombine** near the junction.
- This leaves uncovered fixed ions, which form a thin **depletion region** with no free carriers.
- The ions create an internal electric field, the **barrier potential**: about **0.7 V for Si** and **0.3 V for Ge** (at 25 °C).

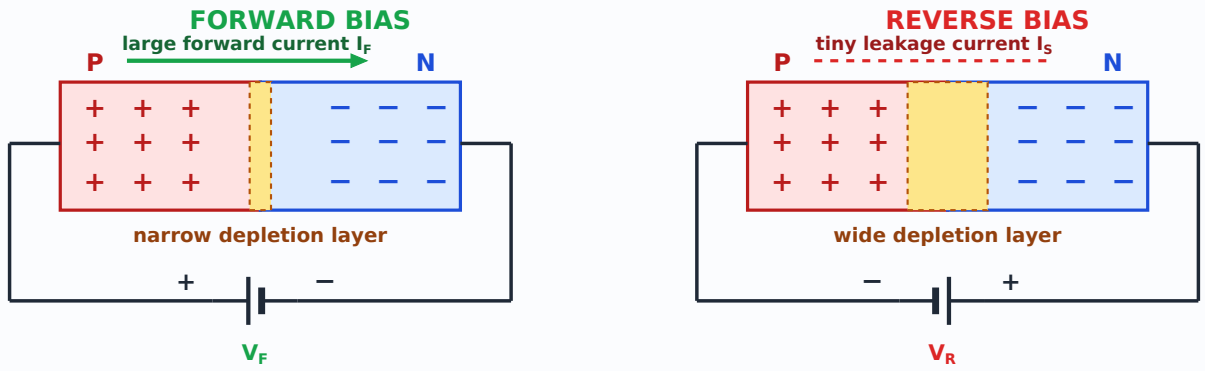


Fig 1.3 PN junction under forward bias (left) and reverse bias (right)

	Forward bias (P to +, N to −)	Reverse bias (P to −, N to +)
Depletion region	Becomes **narrow**	Becomes **wide**
Barrier potential	Reduced	Increased
Current	Large; flows once V > knee voltage	Very small leakage (minority carriers) I_S
Diode acts as	Closed switch (ON)	Open switch (OFF)

3 V-I Characteristics of a PN Junction Diode

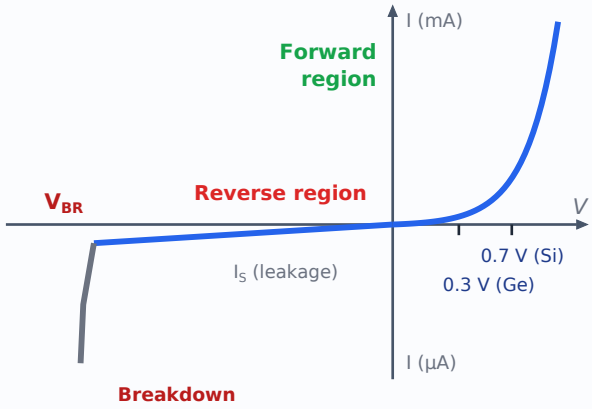


Fig 1.4 V-I characteristic of a PN junction diode (forward and reverse scales differ: mA vs μA)

Diode equation

$$I = I_0 \left(e^{V / \eta V_T} - 1 \right)$$

I_0 = reverse saturation current; $V_T = kT/q \approx 26$ mV at 300 K; η = 1 (Ge), about 2 (Si)

- **Knee (cut-in) voltage:** below it, current is almost zero. Si ≈ 0.7 V, Ge ≈ 0.3 V.
- **Reverse saturation current:** Si in nA, Ge in μA; it rises with temperature (roughly doubles every 10 °C).
- **Breakdown voltage V_{BR}:** a large reverse voltage makes the current rise suddenly and can destroy an ordinary diode.
- **Static resistance** $R_{DC} = V / I$. **Dynamic resistance** $r_d = \Delta V / \Delta I \approx$ **26 mV / I_D** (small in the forward region).

Example: dynamic resistance

A Si diode carries $I_D = 5$ mA. Then $r_d \approx 26$ mV / 5 mA = **5.2 Ω**. At 1 mA it would be 26 Ω, so the diode resists less when it conducts more.

APPLICATIONS

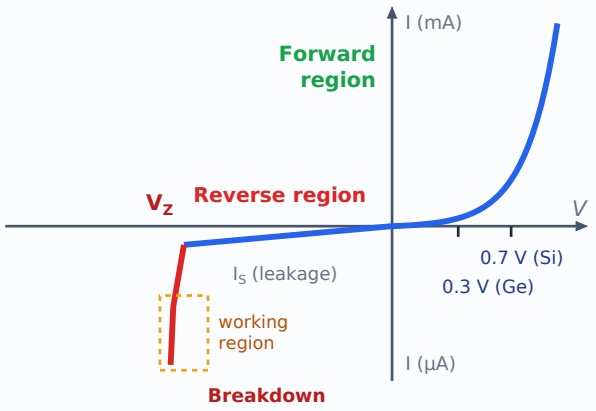
Rectifiers (AC to DC), clippers and clampers, signal detectors, logic gates, reverse-polarity protection, voltage multipliers.

4 Zener Effect and Zener Diode

Zener diode
A specially **heavily doped** PN diode designed to work **safely in the reverse breakdown region**. In breakdown its voltage stays almost **constant (V_Z)** while the current changes widely, so it is a voltage reference/regulator.

Two breakdown mechanisms

	Zener effect	Avalanche effect
Occurs at	Low voltage (V_Z below ~5 V)	Higher voltage (above ~7 V)
Junction	Very heavy doping, very thin depletion layer	Lightly doped, wider depletion layer
Cause	Strong electric field (~10^7 V/m) pulls electrons straight out of covalent bonds (*field emission*)	Fast minority carriers knock out more electrons by *collision*, causing a chain (multiplication)
Temperature coefficient	Negative	Positive



Parameter	Meaning
V_Z	Zener (breakdown) voltage
$I_{Z(min)}$	Minimum current to stay in breakdown
$I_{Z(max)}$	Maximum safe current
$P_{Z(max)}$	$V_Z \times I_{Z(max)}$ (power rating)
Z_Z	Dynamic impedance $\Delta V_Z/\Delta I_Z$ (few Ω)

Fig 1.5 Zener diode characteristic and its main parameters

- **Forward bias:** behaves like an ordinary diode. **Reverse bias:** acts as an open circuit until V_Z, then conducts and holds V_Z.
- **Symbol:** diode symbol with the cathode bar bent like a "Z". Always use a **series resistor** to limit the current: $R_S = (V_{in} - V_Z) / (I_Z + I_L)$.

Example: series resistor for a Zener
Input = 12 V, V_Z = 5.1 V, load current I_L = 20 mA, minimum Zener current I_Z = 5 mA. R_S = (12 − 5.1) / (20 + 5) mA = 6.9 / 25 mA = **276 Ω** (use the nearest standard value, 270 Ω).

APPLICATIONS
Voltage regulators, reference voltage sources, over-voltage protection, waveform clipping, meter protection.

5 Bipolar Junction Transistor (BJT)

DEFINITION
A BJT is a **three-layer, two-junction, three-terminal** device (Emitter, Base, Collector) in which a **small base current controls a large collector current**. "Bipolar" because both electrons and holes take part in conduction.

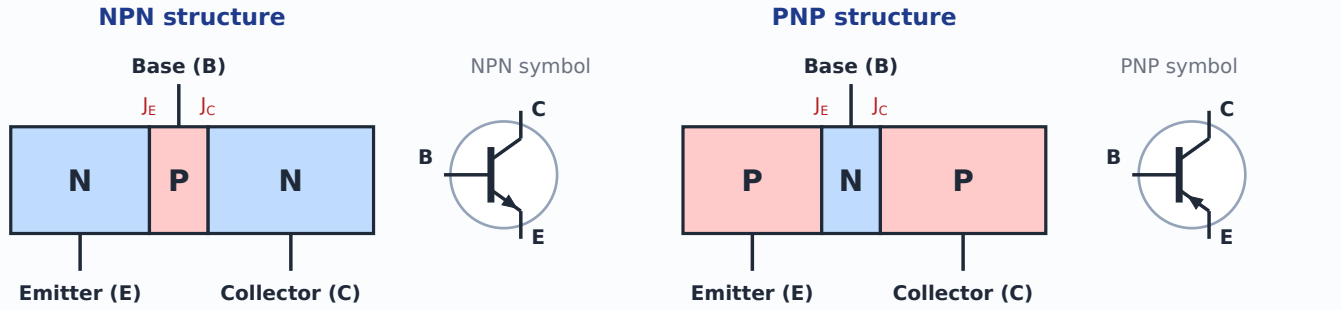


Fig 1.6 Structure and circuit symbols of NPN and PNP transistors (arrow shows direction of conventional emitter current)

Region	Doping	Size	Job
Emitter	Heavy	Medium	Emits (injects) majority carriers
Base	Light	**Very thin**	Controls the carrier flow
Collector	Moderate	**Largest**	Collects the carriers, dissipates heat

Working of an NPN transistor (active mode)

- Emitter-base junction is **forward biased**; collector-base junction is **reverse biased**.
- Electrons from the emitter enter the thin base; only ~1–5 % recombine (this is the small base current I_B).
- The rest are swept into the collector by the reverse field, giving a large $\mathbf{I_C}$.

Mode	EB junction	CB junction	Use
Cut-off	Reverse	Reverse	Switch OFF
Active	**Forward**	**Reverse**	**Amplifier**
Saturation	Forward	Forward	Switch ON

Current relations

$\mathbf{I_E = I_B + I_C}$

$\mathbf{\alpha = I_C / I_E}$ (0.95–0.99) | $\mathbf{\beta = I_C / I_B}$ (20–500)

$\mathbf{\beta = \alpha / (1 - \alpha)}$ | $\mathbf{\alpha = \beta / (1 + \beta)}$ | $\mathbf{I_E = (1 + \beta)\, I_B}$

Example

I_E = 10 mA and I_B = 0.2 mA. Then I_C = 10 − 0.2 = **9.8 mA**, α = 9.8/10 = **0.98**, β = 9.8/0.2 = **49** (check: 0.98/0.02 = 49 ✔).

6 Transistor Configurations: CB, CE and CC

A transistor has three terminals but a two-port network needs four, so **one terminal is made common** to input and output. The common terminal names the configuration.

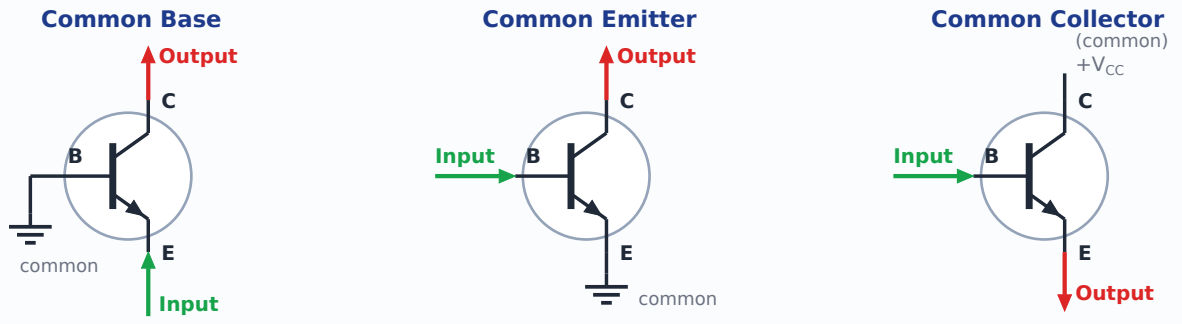


Fig 1.7 Three configurations (green = input, red = output, common terminal is grounded)

Property	CB	CE	CC (emitter follower)
Input terminal / Output	Emitter / Collector	Base / Collector	Base / Emitter
Input resistance	Very low (~50 Ω)	Medium (~1 kΩ)	Very high (~500 kΩ)
Output resistance	Very high (~1 MΩ)	High (~50 kΩ)	Very low (~50 Ω)
Current gain	α (just below 1)	β (high)	1 + β (high)
Voltage gain	High	High	Approx. 1 (slightly below)
Power gain	Medium	**Highest**	Medium
Phase (in to out)	0° (in phase)	**180° (inverted)**	0° (in phase)
Typical use	High-frequency / RF amplifiers	Audio amplifiers, switching	Buffer, impedance matching

EXAM TIP
CE is the most popular configuration because it gives high voltage, current *and* power gain. CC has gain ≈ 1 but is a perfect "buffer" between a high-impedance source and a low-impedance load.

7 Input and Output Characteristics

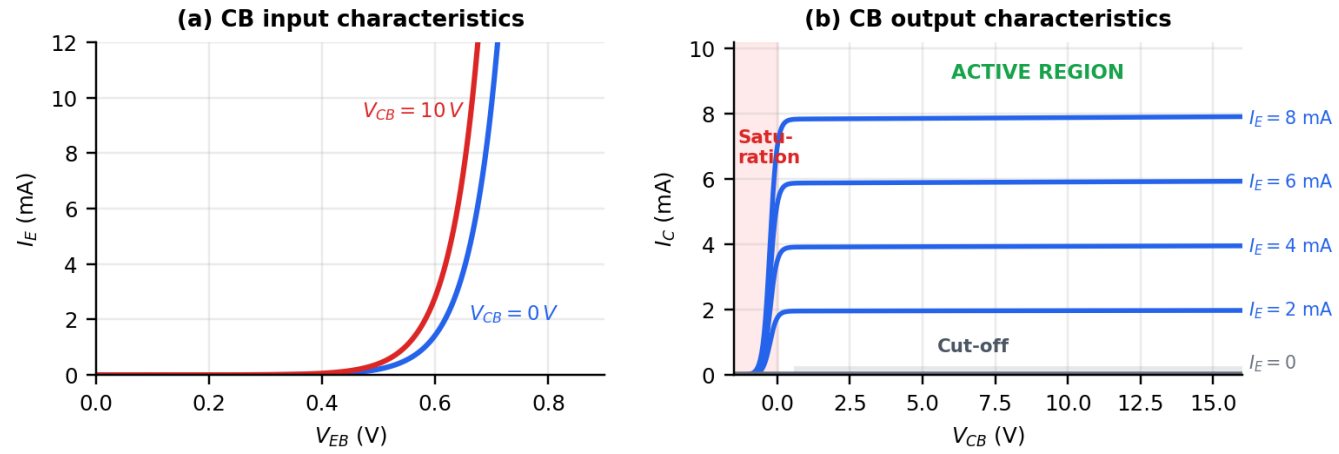


Fig 1.8(a) Common-base input and output characteristics

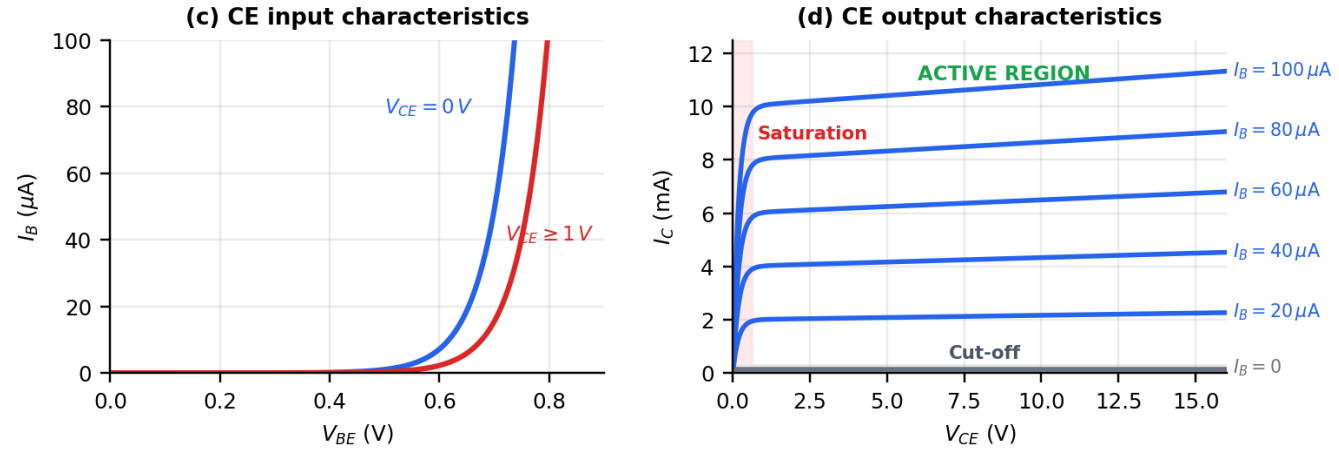


Fig 1.8(b) Common-emitter input and output characteristics

- **CB input (a):** I_E vs V_{EB} at constant V_{CB}. It looks like a forward diode curve; a larger V_{CB} shifts the curve slightly left.
- **CB output (b):** I_C vs V_{CB} at constant I_E. Almost flat lines, because $I_C \approx \alpha I_E$; the curve drops to zero only when V_{CB} is made forward (saturation).
- **CE input (c):** I_B vs V_{BE} at constant V_{CE}; a diode-like curve with a slightly higher cut-in at larger V_{CE}.

- **CE output (d):** I_C vs V_{CE} at constant I_B; $I_C \approx \beta I_B$ and the curves have a gentle upward slope.
- **CC characteristics:** input is I_B vs V_{BC}; output is I_E vs V_{EC}, which is nearly the same as the CE output (since $I_E \approx I_C$).

Region	CE condition	Meaning
Cut-off	$I_B = 0$, $I_C = I_{CEO}$ (tiny)	Transistor OFF; $V_{CE} \approx V_{CC}$
Active	EB forward, CB reverse	Linear amplification; $I_C = \beta I_B$
Saturation	$V_{CE(sat)} \approx 0.2$ V	Transistor fully ON; I_C is set by the external circuit

8 Small-Signal CE Amplifier (Elementary Treatment)

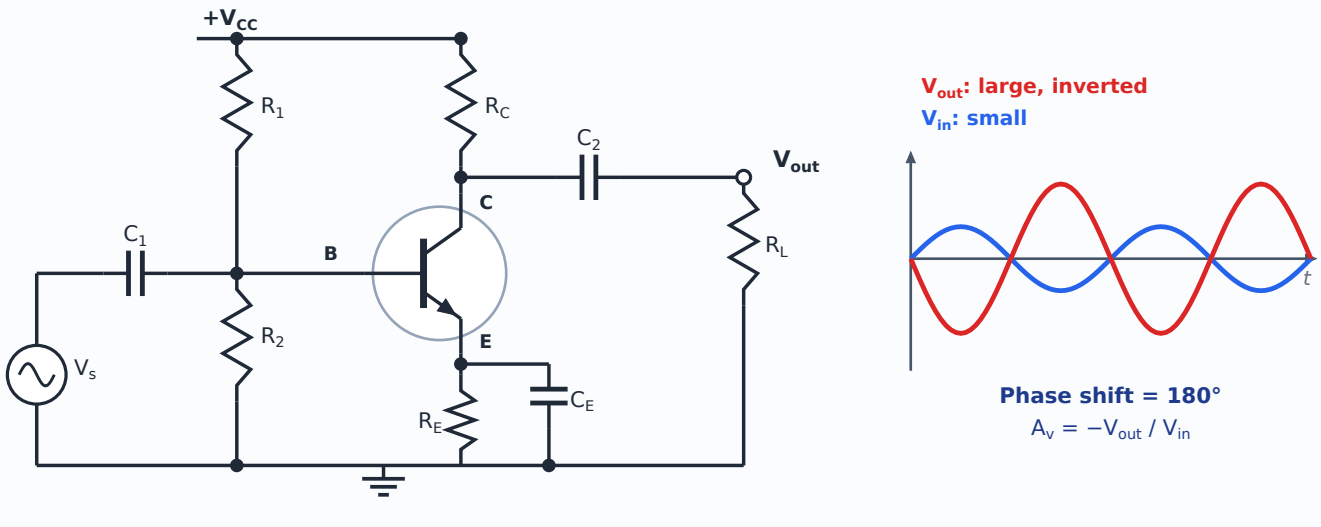


Fig 1.9 Voltage-divider-biased CE amplifier and its input/output waveforms

Role of each component

Component	Purpose
R_1, R_2	Voltage divider that sets a **stable DC bias** (Q-point) at the base
R_C	Converts the collector current change into an output voltage
R_E	Gives **stability** against temperature and β changes (negative feedback)
C_E (bypass)	Short-circuits R_E for AC so the gain is not reduced
C_1, C_2 (coupling)	Pass the AC signal and **block DC** between source, amplifier and load

Working

- **DC:** the biasing network fixes the **Q-point** in the middle of the active region.
- **AC:** in the +ve half-cycle of V_{in}, I_B and I_C increase, so the drop across R_C increases and V_{CE} **decreases**. In the −ve half-cycle the opposite happens.
- So the output is an **amplified and 180° phase-inverted** copy of the input.

Elementary small-signal analysis

DC: $V_B = V_{CC} \cdot R_2/(R_1+R_2)$; $V_E = V_B - 0.7$; $I_C \approx I_E = V_E/R_E$; $V_{CE} = V_{CC} - I_C(R_C+R_E)$

AC: $r_e = 26\text{ mV} / I_E$; $\mathbf{A_v = -(R_C \parallel R_L) / r_e}$; $Z_{in} \approx R_1 \| R_2 \| \beta r_e$; $Z_{out} \approx R_C$

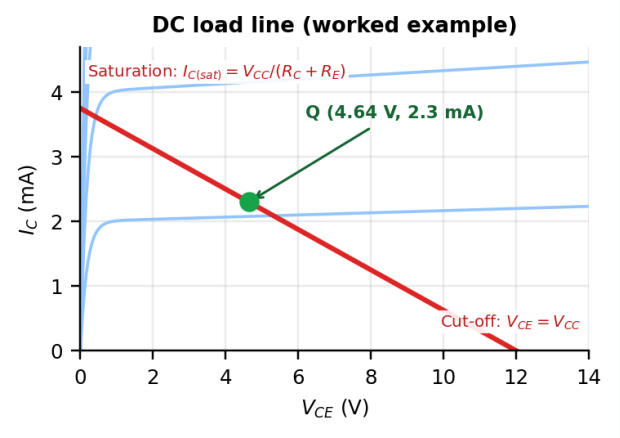
Worked example

$V_{CC} = 12$ V, $R_1 = 30$ kΩ, $R_2 = 10$ kΩ, $R_E = 1$ kΩ, $R_C = 2.2$ kΩ, $R_L = 10$ kΩ.

$V_B = 12 \times 10/40 = 3$ V → $V_E = 2.3$ V → $I_E \approx I_C = 2.3$ mA → $V_{CE} = 12 - 2.3 \times 3.2 =$ **4.64 V** (active ✔).

$r_e = 26/2.3 = 11.3$ Ω; $R_C \| R_L = 1.8$ kΩ; $\mathbf{A_v \approx -1800/11.3 \approx -160}$.

DC load line and Q-point



The **DC load line** joins the two extreme points of the circuit: **cut-off** ($I_C = 0$, $V_{CE} = V_{CC}$) and **saturation** ($V_{CE} \approx 0$, $I_C = V_{CC}/(R_C+R_E)$).

The **Q-point** (quiescent point) is the DC operating point on this line. It must sit near the **middle of the active region** so the signal swings equally up and down without clipping.

Q too high → *saturation clipping*; Q too low → *cut-off clipping*.

Fig 1.10 DC load line and Q-point for the worked example

Transistor as a switch

- **Cut-off** ($I_B = 0$): switch OPEN, $V_{CE} \approx V_{CC}$. • **Saturation** (large I_B): switch CLOSED, $V_{CE(sat)} \approx 0.2$ V.
- Used as the basic element of logic gates, relay/LED drivers and memory cells (links to Unit III).

9 Quick Summary and Exam Corner

Unit I in one glance

- Vacuum tube → transistor (1947) → IC (1958) → VLSI → nano-electronics: smaller, faster, cheaper, cooler.
- PN diode: forward = ON (knee 0.7 V Si, 0.3 V Ge); reverse = OFF with tiny leakage; breakdown at V_{BR}.
- Zener diode: operated in reverse breakdown, holds a constant V_Z; Zener effect (low V) and avalanche (high V).
- BJT: $I_E = I_B + I_C$; $\beta = \alpha/(1-\alpha)$. Modes: cut-off, active (amplifier), saturation.
- CE: highest power gain and 180° phase shift; CB: low R_{in}, good at high frequency; CC: gain ≈ 1, buffer.
- CE amplifier: R_1R_2 bias, R_E stability, C_E bypass, C_1C_2 coupling; $A_v = -R_C \| R_L / r_e$.

Short answer questions (2 marks)

1. What is the barrier potential of Si and Ge diodes?
2. Define dynamic resistance of a diode.
3. Distinguish Zener and avalanche breakdown (any two points).
4. Why is the base of a transistor made thin and lightly doped?
5. Define α and β and write the relation between them.
6. Why is CE the most widely used configuration?

Long answer questions (5 / 10 marks)

1. Explain the forward and reverse V-I characteristics of a PN junction diode with a neat diagram.
2. Explain the Zener diode characteristics and the Zener and avalanche breakdown mechanisms.
3. Draw the CE configuration and explain its input and output characteristics, marking the three regions.
4. Compare CB, CE and CC configurations in a table.
5. With a circuit diagram, explain the working of a small-signal CE amplifier. Derive A_v and state the role of each component.
6. Trace the evolution of electronics from vacuum tubes to nano-electronics.

Numerical practice (with answers)

1. A transistor has $I_B = 40$ µA and $\beta = 100$. Find I_C and I_E. *Ans: $I_C = 4$ mA, $I_E = 4.04$ mA*
2. Find β if $\alpha = 0.99$. *Ans: $\beta = 0.99/0.01 = 99$*
3. A Zener diode ($V_Z = 6.2$ V, $P_{Z(max)} = 0.5$ W). Find $I_{Z(max)}$. *Ans: 0.5/6.2 ≈ 80.6 mA*
4. For a CE amplifier, $I_E = 1.3$ mA, $R_C \| R_L = 2$ kΩ. Find A_v. *Ans: $r_e = 20\ \Omega$, $A_v = -100$*

Common mistakes to avoid

- Mixing up the biasing: **forward** = P to positive; **reverse** = P to negative.
- Saying “transistor amplifies current only”: CE amplifies **voltage, current and power**.
- Forgetting the **series resistor** with a Zener diode, or forgetting that CE **inverts** the signal.
- Drawing characteristic curves without labelling axes, constant parameter and the three regions.

Quick check: pick the right answer

1. The depletion region **widens** in: (a) forward bias (b) reverse bias (c) both (d) none. *Ans: (b)*
2. The phase shift of a CE amplifier is: (a) 0° (b) 90° (c) 180° (d) 360°. *Ans: (c)*
3. Emitter follower is another name for: (a) CB (b) CE (c) CC. *Ans: (c)*
4. A transistor used as an amplifier operates in the: (a) cut-off (b) active (c) saturation region. *Ans: (b)*
5. Zener breakdown below about 5 V is mainly due to: (a) collision (b) field emission (c) diffusion. *Ans: (b)*
6. Which configuration has the highest input resistance? (a) CB (b) CE (c) CC. *Ans: (c)*

BASIC ELECTRONICS ENGINEERING • PART B

UNIT II

Basic Electronic Circuits & Instrumentation

Rectifiers, filters, regulators, public address system, RC-coupled amplifier and instrumentation

WHAT YOU WILL LEARN

1 Block diagram of a DC power supply
2 Half-wave & full-wave bridge rectifier
3 Capacitor filter
4 Zener voltage regulator
5 Public Address (PA) system
6 RC-coupled CE amplifier & response
7 Electronic instrumentation system
8 Summary & exam questions

1 DC Power Supply: Block Diagram

Most electronic circuits need **steady DC**, but the wall socket gives **AC**. A DC power supply converts one into the other in five simple stages.

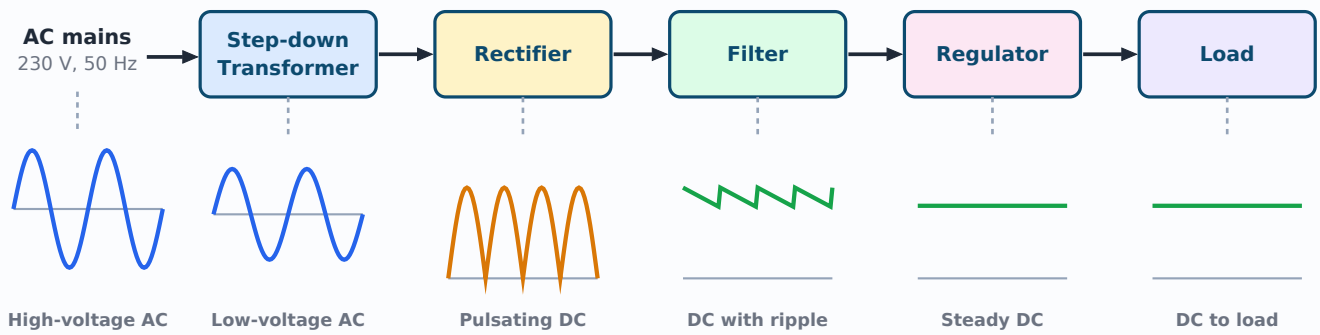


Fig 2.1 Block diagram of a regulated DC power supply, with the waveform at each stage

Block	Function
Step-down transformer	Reduces the 230 V AC mains to a safe, low voltage (e.g. 12 V) and gives electrical isolation
Rectifier	Converts AC to **pulsating DC** using diodes (they conduct in one direction only)
Filter	Smooths the pulsations (removes most of the ripple), usually using a capacitor
Regulator	Keeps the output voltage **constant** despite changes in mains voltage or load (e.g. Zener / IC 7805)
Load	The electronic circuit or device being powered

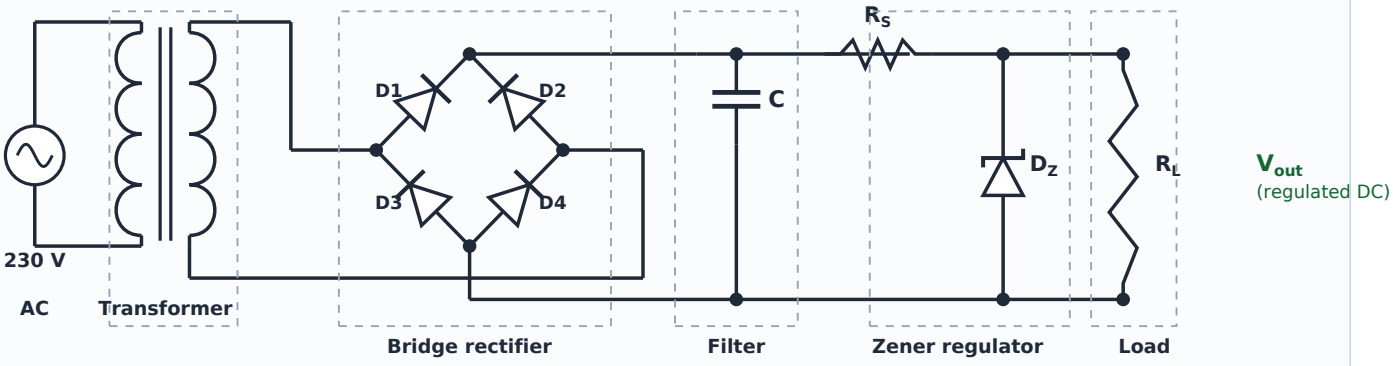


Fig 2.2 Complete regulated power supply circuit (transformer, bridge rectifier, capacitor filter and Zener regulator)

2 Rectifiers: Half-Wave and Full-Wave Bridge

Starting point: the half-wave rectifier

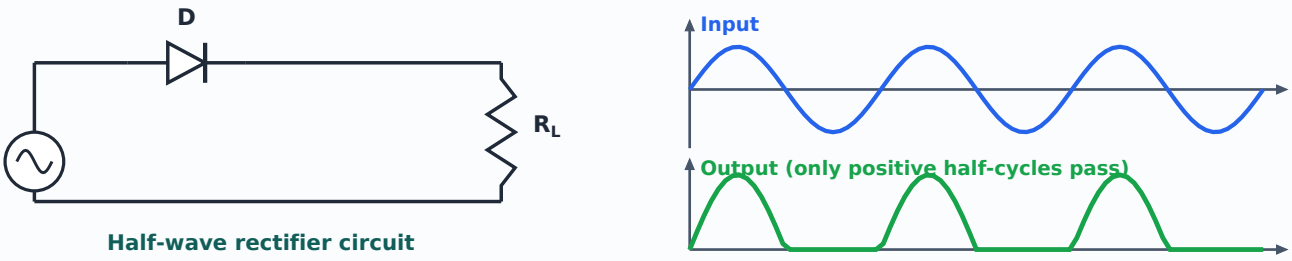


Fig 2.3 Half-wave rectifier circuit and waveforms

- One diode conducts only in the **positive half-cycle**; during the negative half-cycle it is reverse biased and the output is zero.
- $\mathbf{V_{dc} = V_m/\pi \approx 0.318\ V_m}$; ripple factor 1.21; efficiency 40.6 %; PIV = V_m; ripple frequency = f.
- Drawback: half of the input is wasted and the output is very rough, so we use **full-wave** rectification.

Full-wave bridge rectifier

DEFINITION
A **rectifier** converts AC into unidirectional (pulsating DC) current. The **bridge rectifier** uses **four diodes** in a diamond so that **both half-cycles** of the AC input reach the load in the **same direction**, and it needs no centre-tapped transformer.

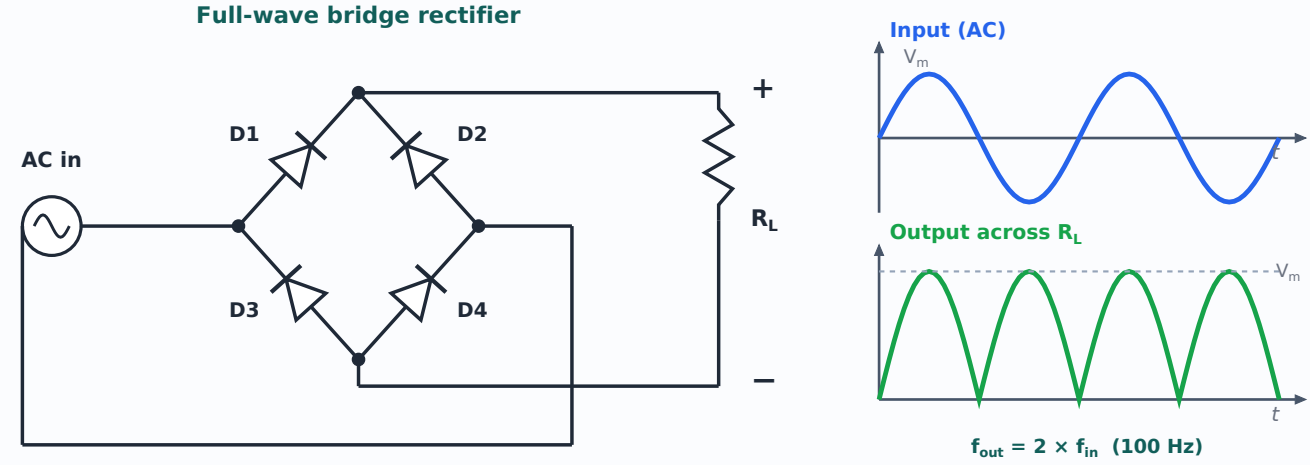


Fig 2.4 Bridge rectifier circuit and its input/output waveforms

Working

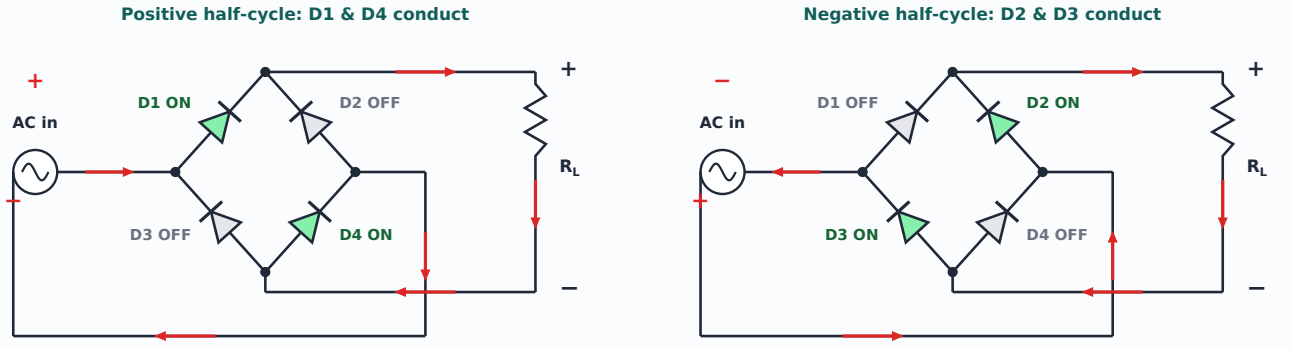


Fig 2.5 Current path in the two half-cycles (red arrows: direction of current; green diodes: conducting)

Input half-cycle	Diodes ON	Diodes OFF	Current through R_L
Positive (top end +)	D_1 and D_4	D_2 and D_3	Top to bottom
Negative (top end −)	D_2 and D_3	D_1 and D_4	**Top to bottom (same direction)**

Important formulas (ideal diodes, peak secondary voltage V_m)
$V_{dc} = 2V_m/\pi \approx 0.637\ V_m$ | $V_{rms} = V_m/\sqrt{2}$ | $I_{dc} = 2I_m/\pi$
Ripple factor $\gamma \approx 0.48$ | Efficiency $\eta = 81.2\ \%$ | PIV $= V_m$ | Ripple frequency $= 2f$

Parameter	Half-wave	Centre-tap full-wave	Bridge full-wave
Number of diodes	1	2	4
V_{dc}	V_m/π	$2V_m/\pi$	$2V_m/\pi$
PIV of each diode	V_m	$2V_m$	**V_m**
Ripple factor	1.21	0.48	0.48
Efficiency	40.6 %	81.2 %	81.2 %
Ripple frequency	f	2f	2f
Centre-tap transformer	No	**Yes**	**No**

Performance term	Meaning
Ripple factor (γ)	Ratio of AC (ripple) component to DC component in the output; smaller is better ($\gamma = V_{r(rms)} / V_{dc}$)
PIV (peak inverse voltage)	Maximum reverse voltage that a diode must withstand without breaking down
Efficiency (η)	DC output power / AC input power × 100 %
Regulation	How little the output voltage changes with load current (and input voltage)

Example
Secondary = 12 V rms. $V_m = 12 \times 1.414 = 16.97$ V. Ideal $V_{dc} = 0.637 \times 16.97 \approx$ **10.8 V**.
In practice two diodes (2 × 0.7 V = 1.4 V) conduct at a time: $V_{dc} \approx 0.637 \times (16.97 - 1.4) \approx$ **9.9 V**. Each diode needs PIV ≥ 17 V (a 1N4007 is ample).

Why the bridge is preferred
- Uses the whole transformer winding all the time (better utilisation), PIV across each diode is only V_m.
- Output ripple is at 2f (100 Hz), which is easier to filter than f (50 Hz) in a half-wave rectifier.
- Disadvantage: two diode drops (≈1.4 V) in the path, so a little more power loss at low voltages.

3 Capacitor Filter

The rectifier output is DC in direction but not steady: it pulses at 100 Hz. A **capacitor filter** (a large electrolytic capacitor connected **in parallel** with the load) stores charge at the peaks and releases it in the gaps.

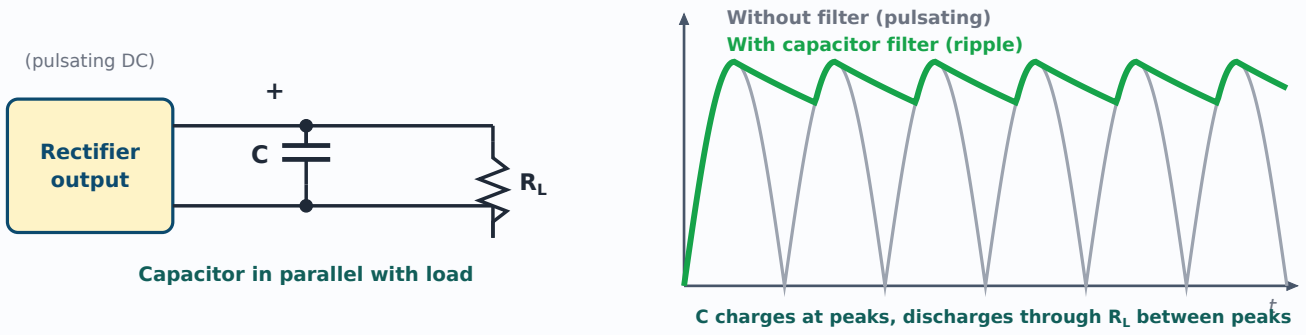


Fig 2.6 Capacitor filter and the smoothed output voltage

- **Rising part:** when the rectified voltage rises above the capacitor voltage, the diodes conduct and the capacitor **charges** quickly to the peak V_m.
- **Falling part:** the input drops below the capacitor voltage, the diodes switch OFF and the capacitor **discharges slowly** through R_L (time constant $R_L C$) so the load voltage barely falls.
- **Next peak:** the capacitor is topped up again, which gives a nearly flat DC with a small **ripple**.

Effects of the capacitor filter

- Larger C or larger R_L (lighter load) → **less ripple** and higher V_{dc} (closer to V_m).
- Heavy load (small R_L) → capacitor drains faster → **more ripple**.
- Typical values: 470 µF to 4700 µF electrolytic; observe correct polarity and a voltage rating above V_m.

4 Simple Zener Voltage Regulator

Even after filtering, the DC output changes when the mains voltage or the load changes. A **regulator** keeps V_{out} constant. The simplest one uses a **Zener diode in reverse breakdown** in parallel with the load, plus a series resistor R_S.

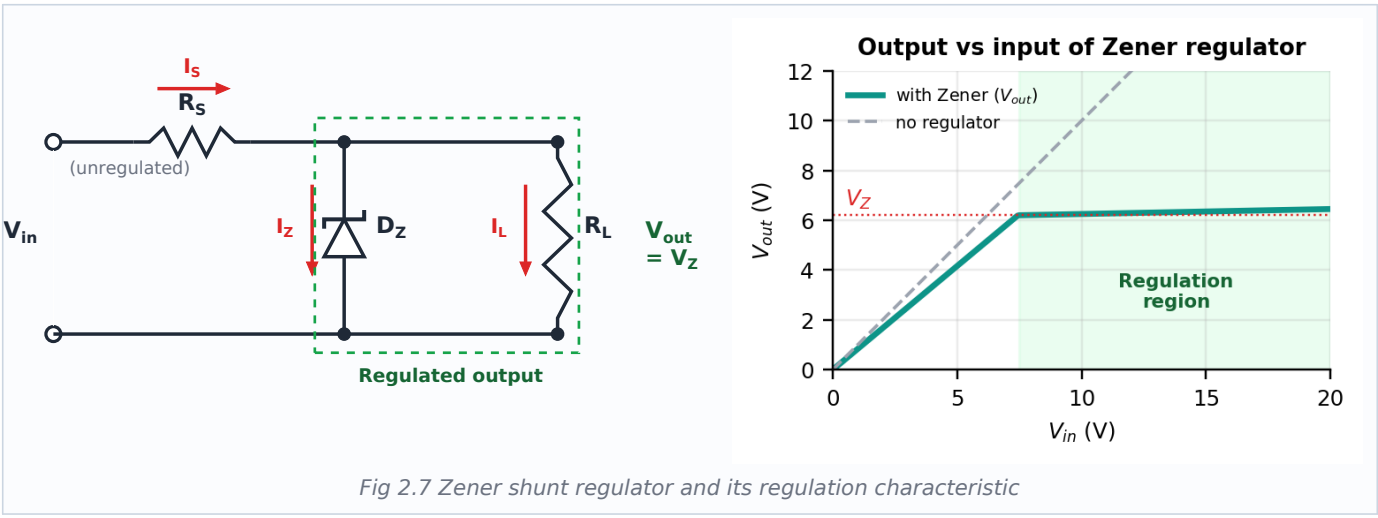


Fig 2.7 Zener shunt regulator and its regulation characteristic

Working

- The Zener diode is **reverse biased** and operates in breakdown, so the voltage across it (and the load) stays at $\mathbf{V_Z}$.
- **If V_{in} increases:** I_S rises; the extra current flows through the Zener (I_Z ↑). The extra drop appears across R_S, so V_{out} stays at V_Z.
- **If load current I_L increases:** I_Z decreases by the same amount, I_S stays almost constant, so V_{out} stays at V_Z.
- **Rule:** I_Z must stay between $I_{Z(min)}$ and $I_{Z(max)}$ for regulation to hold.

Design equations

$\mathbf{I_S = I_Z + I_L = (V_{in} - V_Z) / R_S}$ | $\mathbf{V_{out} = V_Z}$ | $\mathbf{P_{Z(max)} = V_Z\, I_{Z(max)}}$

Line regulation = $\Delta V_{out}/\Delta V_{in} \times 100\ \%$ | Load regulation = $(V_{NL} - V_{FL})/V_{FL} \times 100\ \%$

Example

V_{in} = 12 V, V_Z = 5.1 V, R_L = 1 kΩ → I_L = 5.1 mA. For I_Z = 5 mA: $R_S = (12 - 5.1)/(5 + 5.1)$ mA = **683 Ω ≈ 680 Ω**.

Limitations

Wastes power in R_S and the Zener, fixed output, suitable only for small load currents. Higher currents use transistor or IC regulators (e.g. 7805).

5 Amplifiers and the Public Address (PA) System

DEFINITION

An **amplifier** increases the strength (voltage, current or power) of a signal without changing its shape. Voltage gain $A_v = V_o/V_i$; in decibels, **gain (dB) = 20 $\log_{10} A_v$**.

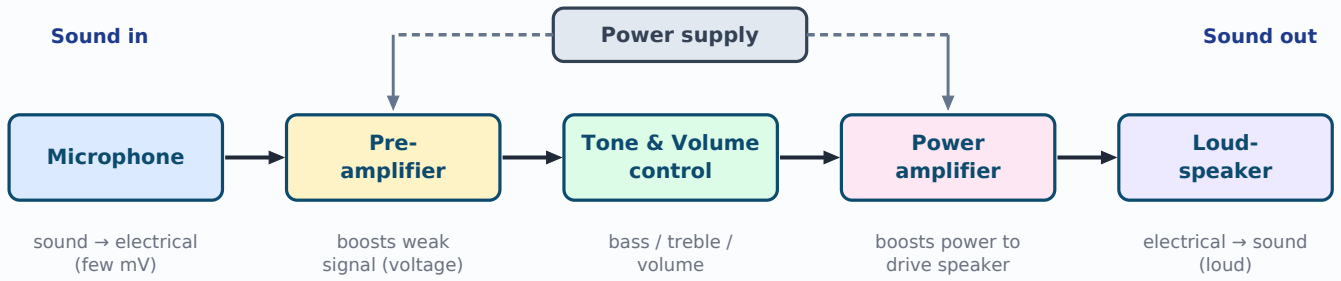


Fig 2.8 Block diagram of a Public Address (PA) system

Block	Role
Microphone	Transducer: converts sound waves into a tiny electrical signal (a few mV)
Pre-amplifier	Low-noise voltage amplifier that raises the weak signal to a usable level
Tone / volume control	Adjusts bass, treble and loudness; shapes the signal before power amplification
Power amplifier	Supplies high current/power needed to drive the loudspeaker
Loudspeaker	Transducer: converts the electrical signal back to sound
Power supply	Gives DC to the amplifier stages

Voltage gain A_v	1	2	10	100	1000	0.707
Gain in dB	0	6 dB	20 dB	40 dB	60 dB	−3 dB

Acoustic feedback (“howling”)

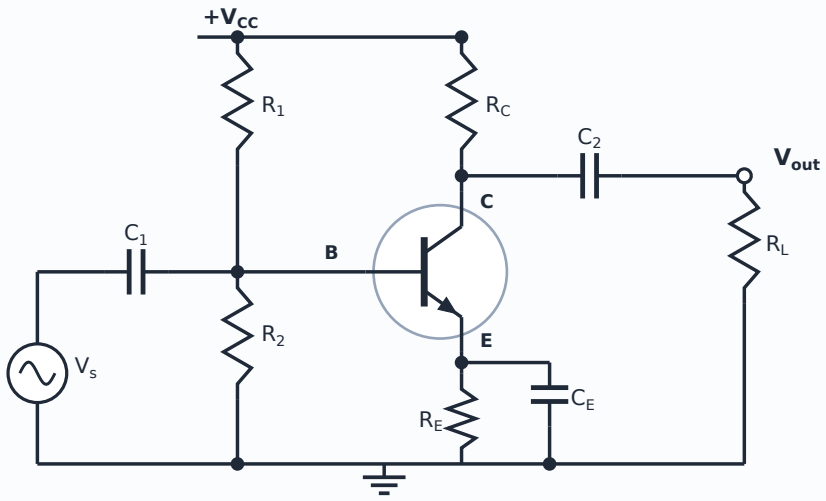
If the loudspeaker sound re-enters the microphone, it is amplified again and again, producing a loud whistle. Cure: keep the microphone away from the speakers, lower the gain or use directional microphones.

APPLICATIONS

Railway stations, airports, auditoriums, classrooms, stadiums, religious places, emergency announcements.

6 RC-Coupled CE Amplifier and its Frequency Response

When one transistor stage cannot give enough gain, stages are cascaded. They are linked by a **coupling network**; the most common is **RC coupling**, where a capacitor passes the AC signal and blocks the DC of the previous stage.



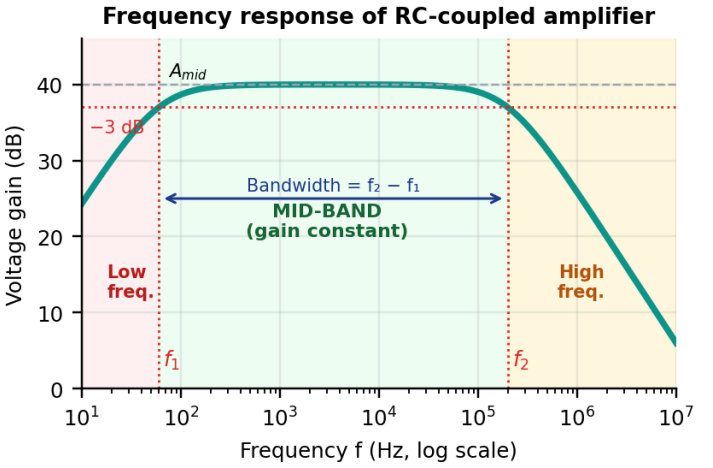
Components

- **R_1, R_2** – voltage-divider bias
- **R_C** – collector (load) resistor
- **R_E** – bias stabilisation
- **C_E** – bypass: prevents gain loss
- **C_1, C_2** – coupling capacitors: pass AC, block DC
- **R_L** – next stage / load

Fig 2.9 Single stage of an RC-coupled common-emitter amplifier

Working

- **DC:** R_1, R_2, R_E fix the Q-point in the active region, so the transistor always conducts.
- **AC:** the input signal passes through C_1 to the base and varies I_B. The collector current varies β times more, producing a large signal voltage across R_C.
- The amplified signal, **180° out of phase** with the input, goes through C_2 to the load. C_2 blocks the DC at the collector.



Why does the gain change with frequency?

Low frequencies: reactance $X_C = 1/(2\pi fC)$ of C_1, C_2 and C_E is large, so part of the signal is dropped across them → gain falls.

Mid frequencies: capacitors behave like short circuits; transistor capacitances are negligible → **gain is maximum and constant**.

High frequencies: transistor junction and stray capacitances shunt the signal, and β falls → gain falls.

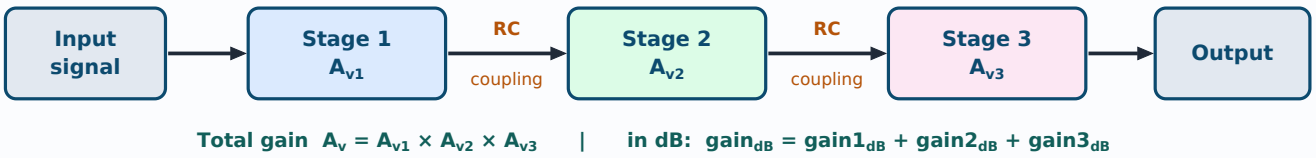
Fig 2.10 Frequency response curve (gain in dB against log frequency)

Bandwidth

The **half-power (−3 dB) frequencies** f_1 (lower) and f_2 (upper) are where the gain drops to **0.707 A_{mid}** ($A_{mid}/\sqrt{2}$). **Bandwidth BW = $f_2 - f_1$**. For audio amplifiers this is about 20 Hz to 20 kHz.

Advantages	Disadvantages
Simple, cheap and compact; no bulky transformers or inductors	Poor impedance matching, so low power transfer to a low-resistance load (e.g. a speaker)
Good, wide, flat response over the audio range	Gain falls at very low and very high frequencies
Little frequency distortion; easy to cascade many stages	Cannot amplify DC; power gain is modest (so used for voltage amplification)

Multistage amplifiers and coupling methods



Total gain $A_v = A_{v1} \times A_{v2} \times A_{v3}$ | **in dB:** $gain_{dB} = gain1_{dB} + gain2_{dB} + gain3_{dB}$

Fig 2.11 Cascaded amplifier stages joined by coupling networks

Coupling	How it works	Remark
RC coupling	Capacitor + resistor pass AC and block DC	Cheap, wide bandwidth: audio voltage amplifiers
Transformer coupling	Transformer transfers the signal and matches impedance	Good power transfer but bulky, costly; narrow band
Direct coupling	Output of one stage connected directly to the next	Can amplify DC; stage biasing interacts (drift)

7 Electronic Instrumentation System

DEFINITION

Electronic instrumentation is the use of electronic circuits to **measure, display, record and control** physical quantities (temperature, pressure, light, force, speed, heart-beat, ...) by first converting them into electrical signals.

Example: digital thermometer → heat ▶ thermistor ▶ amplifier ▶ ADC ▶ numeric display

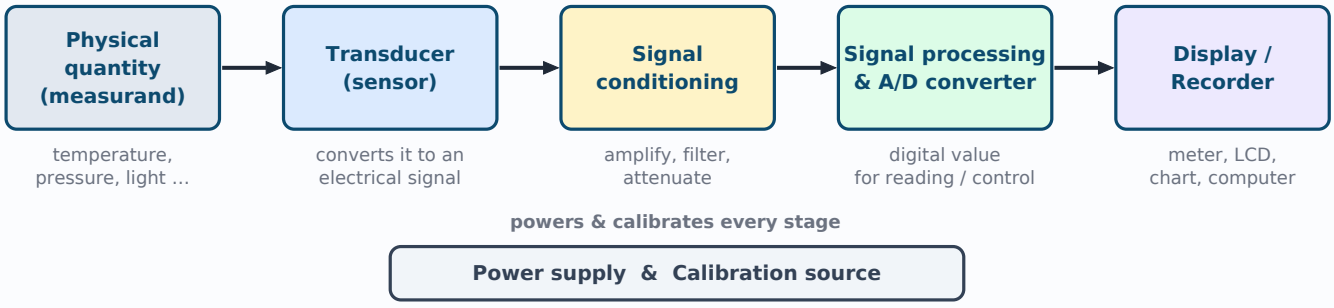


Fig 2.12 Block diagram of an electronic instrumentation system

Block	Function	Example
Transducer / sensor	Converts a physical quantity into an electrical signal	Thermocouple, thermistor, LDR, strain gauge, microphone
Signal conditioning	Amplifies, filters, attenuates or linearises the weak signal	Op-amp amplifier, low-pass filter
Signal processing / ADC	Converts analog to digital; computes, compares, controls	Microcontroller, ADC chip
Display / recorder	Presents the result for the user or stores/transmits it	LCD, LED, meter, chart recorder, PC
Power supply, calibration	Powers all blocks; reference to keep accuracy	Regulated DC supply

Common transducers

Quantity	Transducer	Principle
Temperature	Thermistor, thermocouple, RTD	Resistance / voltage changes with temperature
Light	LDR, photodiode	Resistance / current changes with light intensity
Pressure, force, weight	Strain gauge, load cell	Resistance changes when stretched
Displacement	LVDT, potentiometer	Induced voltage / resistance varies with position
Sound	Microphone	Sound pressure to voltage

Active vs passive transducers

- **Active** transducers generate their own voltage (thermocouple, piezo-electric crystal, photovoltaic cell).
- **Passive** transducers need an external supply (thermistor, strain gauge, LDR).

APPLICATIONS

Digital multimeter, oscilloscope, ECG / patient monitor, weighing machine, industrial process control, weather stations, smart-home sensors.

Advantages of electronic instruments

High sensitivity and speed, remote measurement, easy digital display and storage, small size, and the ability to connect to computers and control systems.

8 Quick Summary and Exam Corner

Unit II in one glance

- DC supply = **Transformer → Rectifier → Filter → Regulator → Load**.
- Bridge rectifier: 4 diodes; two conduct in each half-cycle; $V_{dc} = 2V_m/\pi$; PIV = V_m; ripple frequency = 2f; $\eta = 81.2$ %.
- Capacitor filter: C in parallel with load; charges at peaks, discharges slowly; larger C → smaller ripple.
- Zener regulator: reverse-biased Zener in parallel with load plus R_S; $V_{out} = V_Z$; $R_S = (V_{in} - V_Z)/(I_Z + I_L)$.
- PA system: Microphone → Pre-amp → Tone control → Power amp → Loudspeaker.
- RC-coupled CE amplifier: flat mid-band; gain falls at low f (coupling/bypass caps) and high f (junction/stray caps); BW = $f_2 - f_1$.
- Instrumentation: Measurand → Transducer → Signal conditioning → Processing/ADC → Display.

Short answer questions (2 marks)

1. List the blocks of a regulated DC power supply.
2. Why is a bridge rectifier preferred over a centre-tap rectifier?
3. What is the ripple frequency of a full-wave rectifier with 50 Hz input?
4. State the function of a filter and of a regulator.
5. What is the purpose of the bypass capacitor in a CE amplifier?
6. Define bandwidth of an amplifier.
7. What is a transducer? Give two examples.

Long answer questions (5 / 10 marks)

1. Draw the block diagram of a DC power supply and explain each block with waveforms.
2. With a neat circuit diagram, explain the working of a full-wave bridge rectifier. Give its V_{dc}, ripple factor, efficiency and PIV.
3. Explain the working of a capacitor filter with waveforms.
4. Explain the working of a Zener voltage regulator for variations in input voltage and load.
5. Draw the block diagram of a PA system and explain the function of each block.
6. Draw the circuit of an RC-coupled CE amplifier, explain its working and sketch its frequency response.

7. Draw and explain the block diagram of an electronic instrumentation system.

Numerical practice (with answers)

1. A bridge rectifier has a secondary of 20 V rms. Find V_m, ideal V_{dc} and PIV. *Ans: V_m = 28.3 V, V_{dc} = 18.0 V, PIV = 28.3 V*
2. A Zener (V_Z = 9.1 V) is fed from 15 V through R_S = 590 Ω. With no load, find the Zener current. *Ans: (15 − 9.1)/590 = 10 mA*
3. An amplifier has A_{mid} = 200, f_1 = 40 Hz, f_2 = 80 kHz. Find the bandwidth and the gain at the −3 dB points. *Ans: BW ≈ 79.96 kHz; gain = 200/√2 ≈ 141*
4. Convert a voltage gain of 100 into dB. *Ans: 20 log 100 = 40 dB*

BASIC ELECTRONICS ENGINEERING • PART B

UNIT III

Digital Electronics

Number systems, codes, logic gates, Boolean algebra, adders, flip-flops, registers and counters

WHAT YOU WILL LEARN

1 Number systems & conversions
2 BCD, Excess-3, Gray & Hamming codes
3 Logic gates & universal gates
4 Boolean algebra & theorems
5 Half adder & full adder
6 Sequential circuits & flip-flops
7 Registers and counters
8 Summary & exam questions

1 Overview of Number Systems

Digital circuits work with only two voltage levels, **HIGH (1)** and **LOW (0)**, so they naturally use the **binary** number system. Octal and hexadecimal are shorthand ways to write long binary numbers.

System	Base	Digits used	Example
Decimal	10	0–9	$(45)_{10}$
Binary	2	0, 1	$(101101)_{2}$
Octal	8	0–7	$(55)_{8}$
Hexadecimal	16	0–9, A–F (A=10 … F=15)	$(2D)_{16}$

Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Bin	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Oct	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Conversion methods

Conversion	Method	Example
Decimal → Binary	Divide repeatedly by 2; read remainders **bottom to top**	45 → 22 r1, 11 r0, 5 r1, 2 r1, 1 r0, 0 r1 → **101101**
Binary → Decimal	Add the weights (…8, 4, 2, 1 . ½, ¼, ⅛)	1101.101 = 8+4+1+0.5+0.125 = **13.625**
Binary → Octal	Group bits in **3s** from the binary point	101 110 011 → $\mathbf{563}_{8}$
Binary → Hex	Group bits in **4s**	1011 1100 → $\mathbf{BC}_{16}$
Hex → Decimal	Multiply digits by powers of 16	2F = 2×16 + 15 = **47**
Decimal → Hex	Divide repeatedly by 16	255 → F, F → **FF**

Binary arithmetic

Addition: 0+0 = 0, 0+1 = 1, 1+0 = 1, **1+1 = 10** (sum 0, carry 1). **Subtraction** is done by adding the **2's complement**.

1's complement = flip every bit; **2's complement** = 1's complement + 1.

Example (4-bit) 7 − 5: 0111 + 2's comp(0101) = 0111 + 1011 = 1 0010 → discard the carry → **0010 = 2** ✓

2 Digital Codes: BCD, Excess-3, Gray and Hamming

A **code** represents numbers or characters as groups of bits. Different codes are chosen for ease of calculation, error reduction or error correction.

Decimal	Binary	BCD (8421)	Excess-3	Gray
0	0000	0000	0011	0000
1	0001	0001	0100	0001
2	0010	0010	0101	0011
3	0011	0011	0110	0010
4	0100	0100	0111	0110
5	0101	0101	1000	0111
6	0110	0110	1001	0101
7	0111	0111	1010	0100
8	1000	1000	1011	1100
9	1001	1001	1100	1101

BCD (Binary Coded Decimal)

Each decimal **digit** is written separately as a 4-bit binary number (weights 8-4-2-1). Codes 1010–1111 are **invalid**.

Example: $(47)_{10}$ = **0100 0111**$_{BCD}$. Used in digital clocks, calculators and meters.

Excess-3 code

BCD + 0011 (3) for every digit. It is **self-complementing** (9's complement = flip all bits), useful in decimal arithmetic.

Example: 5 → 0101 + 0011 = **1000**.

Gray code (reflected binary)

Only one bit changes between consecutive numbers, which avoids glitches in encoders and position sensors.

Binary → Gray: keep the MSB; each next Gray bit = XOR of that binary bit and the one before it. 1011 → 1, 1⊕0=1, 0⊕1=1, 1⊕1=0 → **1110**.

Gray → Binary: keep the MSB; each next binary bit = previous binary bit ⊕ current Gray bit. 1110 → 1, 1⊕1=0, 0⊕1=1, 1⊕0=1 → **1011**.

Hamming code (error detection and correction)

- A **Hamming (7,4)** code adds **3 parity bits** to 4 data bits so that a **single-bit error** can be found *and corrected*.
- Bit positions 1 to 7: parity bits sit at positions **1, 2, 4** (powers of 2); data bits at 3, 5, 6, 7.
- Each parity bit checks a group (even parity): **P1** → positions 1,3,5,7; **P2** → 2,3,6,7; **P4** → 4,5,6,7.

Step	Working for data 1011 (even parity)
Place data	Positions 3,5,6,7 = **1, 0, 1, 1**
P1 (1,3,5,7)	bits 3,5,7 = 1,0,1 → two 1s → **P1 = 0**
P2 (2,3,6,7)	bits 3,6,7 = 1,1,1 → three 1s → **P2 = 1**
P4 (4,5,6,7)	bits 5,6,7 = 0,1,1 → two 1s → **P4 = 0**
Code word	positions 1…7 → **0 1 1 0 0 1 1**
Error check	If received 0110**1**11 (bit 5 flipped): C1 = 1, C2 = 0, C4 = 1 → C4C2C1 = 101 = 5 → **bit 5 is wrong; flip it**

3 Logic Gates and Universal Gates

A **logic gate** is the basic building block of digital circuits: it takes one or more binary inputs and produces one binary output according to a logical rule.

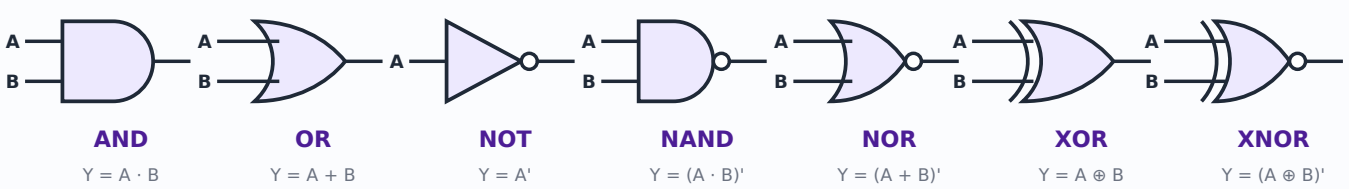


Fig 3.1 Symbols and Boolean expressions of the basic logic gates

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Gate	Functionality (in words)
NOT (inverter, 1 input: 0→1, 1→0)	Output is the **opposite** of the input
AND	Output is 1 **only when all** inputs are 1
OR	Output is 1 when **at least one** input is 1
NAND / **NOR**	AND / OR followed by NOT (output is the complement)
XOR	Output is 1 when the inputs are **different** (odd number of 1s)
XNOR	Output is 1 when the inputs are the **same** (equality detector)

Universal gates
NAND and **NOR** are called **universal gates** because **any** logic function (NOT, AND, OR and hence all circuits) can be built using only NAND gates or only NOR gates. They are cheap, fast and easy to fabricate.

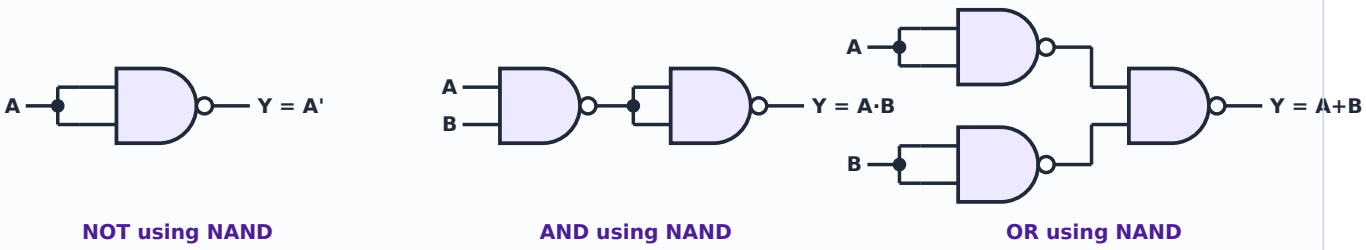


Fig 3.2 NOT, AND and OR gates built using only NAND gates

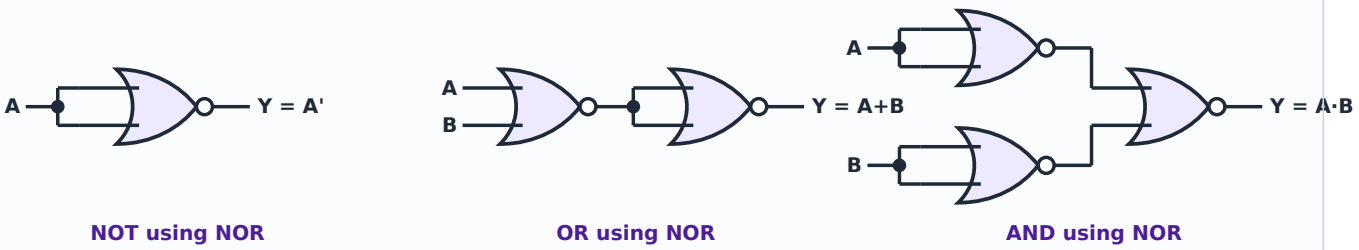


Fig 3.3 NOT, OR and AND gates built using only NOR gates

4 Boolean Algebra: Properties and Theorems

Boolean algebra is the algebra of two-valued variables (0 and 1) with the operations AND (·), OR (+) and NOT (A′ means "NOT A"). It is used to **simplify** logic circuits.

Law	AND form	OR form
Identity	A · 1 = A	A + 0 = A
Null (domination)	A · 0 = 0	A + 1 = 1
Idempotent	A · A = A	A + A = A
Complement	A · A′ = 0	A + A′ = 1
Involution	(A′)′ = A	(A′)′ = A
Commutative	A · B = B · A	A + B = B + A
Associative	(A·B)·C = A·(B·C)	(A+B)+C = A+(B+C)
Distributive	A·(B + C) = AB + AC	A + BC = (A + B)(A + C)
Absorption	A · (A + B) = A	A + A·B = A
Useful	A · (A′ + B) = AB	A + A′B = A + B
De Morgan's	**(A · B)′ = A′ + B′**	**(A + B)′ = A′ · B′**

Duality principle
Every Boolean identity stays true if we swap **AND ↔ OR** and **0 ↔ 1**. (That is why each law above has two forms.)

De Morgan's theorems in words

- The complement of a product = the sum of the complements. • The complement of a sum = the product of the complements.
- In circuits: NAND = OR of inverted inputs; NOR = AND of inverted inputs (bubbled gates).

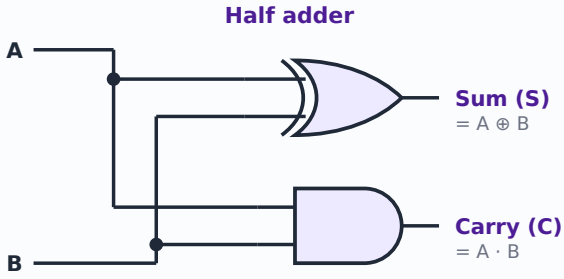
A	B	(A·B)′	A′ + B′	(A + B)′	A′ · B′
0	0	1	1	1	1
0	1	1	1	0	0
1	0	1	1	0	0
1	1	0	0	0	0

Simplification example
Simplify Y = A′B + AB′ + AB.
Y = A′B + A(B′ + B) = A′B + A·1 = A′B + A = **A + B** (using A + A′B = A + B). So the three-term circuit is just one OR gate.

5 Simple Combinational Circuits: Half and Full Adder

DEFINITION
A **combinational circuit** has outputs that depend **only on the present inputs** (no memory). Examples: adders, subtractors, multiplexers, decoders, encoders.

Half adder: adds two 1-bit numbers



A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Fig 3.4 Half adder: logic diagram and truth table (Sum = A ⊕ B, Carry = A · B)

Full adder: adds two bits and a carry-in

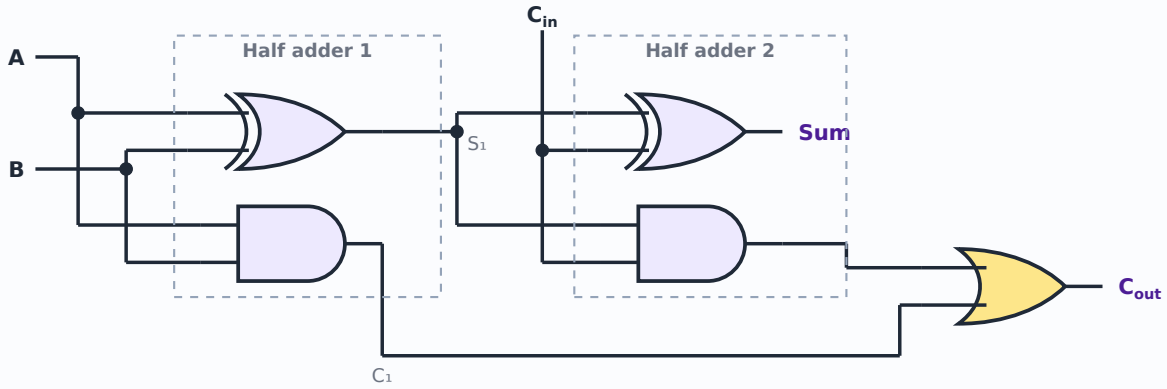


Fig 3.5 Full adder built from two half adders and an OR gate

A	B	C_{in}	Sum	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Adder equations

Half adder: $S = A \oplus B$, $C = A \cdot B$

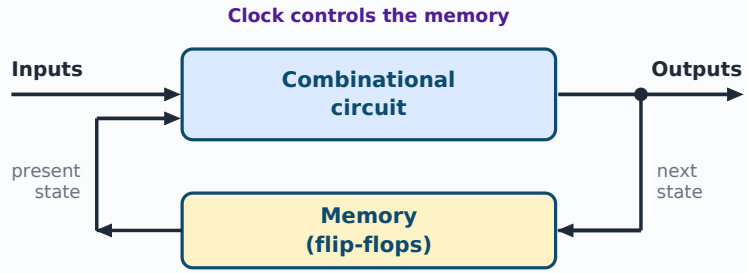
Full adder: $Sum = A \oplus B \oplus C_{in}$, $C_{out} = A{\cdot}B + C_{in}{\cdot}(A \oplus B)$

APPLICATIONS

Arithmetic Logic Unit (ALU) of every processor, calculators, address calculation, counters and digital signal processing. Cascading full adders gives an *n*-bit parallel adder.

6 Introduction to Sequential Circuits and Flip-Flops

	Combinational circuit	Sequential circuit
Output depends on	Present inputs only	Present inputs **and past state**
Memory element	None	Flip-flops / latches
Clock	Not needed	Usually clocked
Examples	Adder, decoder, multiplexer	Registers, counters, memories

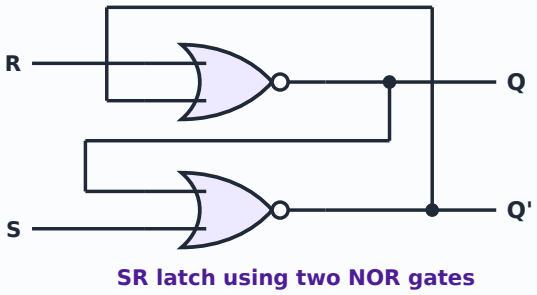


Flip-flop = a 1-bit memory cell with two stable states (Q = 0 or Q = 1). It remembers its state until the inputs and clock change it.

Latch responds to inputs at any time (level sensitive); a **flip-flop** changes only at the clock edge (edge triggered).

A flip-flop has outputs **Q** and its complement **Q′**.

Fig 3.6 Block diagram of a sequential circuit



Working of the NOR latch

S = 1, R = 0 → **Q = 1** (set)

S = 0, R = 1 → **Q = 0** (reset)

S = 0, R = 0 → **holds** the previous state (memory)

S = 1, R = 1 → **forbidden** (both outputs 0)

Fig 3.7 SR latch: the basic memory element

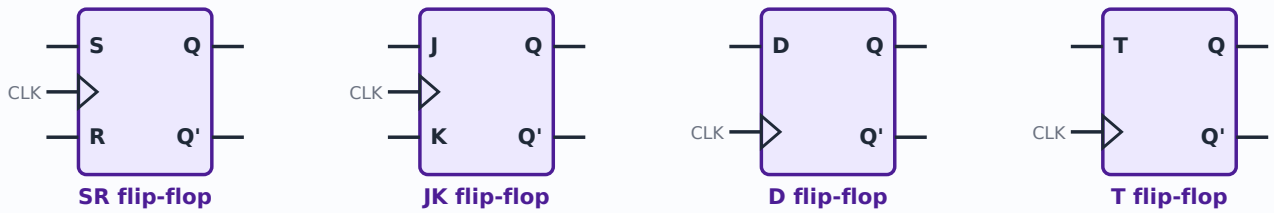


Fig 3.8 Block symbols of SR, JK, D and T flip-flops (triangle = clock input)

S	R	Q^+
0	0	Q
0	1	0
1	0	1
1	1	X

J	K	Q^+
0	0	Q
0	1	0
1	0	1
1	1	Q′

D	Q^+
0	0
1	1

T	Q^+
0	Q
1	Q′

Fig 3.9 Characteristic tables: SR, JK, D and T (Q^+ = next state; X = not allowed)

Flip-flop	Behaviour	Characteristic equation	Remark
SR	Set / Reset / Hold	$Q^+ = S + R'Q$	S = R = 1 is invalid
JK	Like SR, but 11 → **toggle**	$Q^+ = JQ' + K'Q$	Universal flip-flop; race-around in level-triggered form (cured by master-slave / edge triggering)
D	Output follows D after the clock (delay)	$Q^+ = D$	Used in registers and data storage
T	Toggles when T = 1	$Q^+ = T \oplus Q$	JK with J = K = T; used in counters

7 Registers and Counters (Elementary Treatment)

Register
A group of flip-flops that stores a binary word. An *n*-bit register uses *n* flip-flops. A **shift register** also moves the stored bits left or right on every clock pulse.

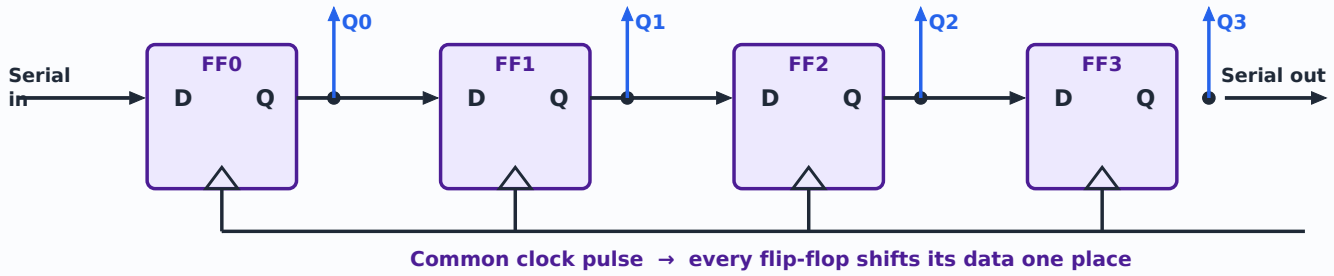


Fig 3.10 4-bit shift register (serial-in, parallel/serial-out) using D flip-flops

Type	Data in	Data out	Typical use
SISO	Serial	Serial	Time delay of digital signals
SIPO	Serial	Parallel	Serial-to-parallel conversion (receiver)
PISO	Parallel	Serial	Parallel-to-serial conversion (transmitter)
PIPO	Parallel	Parallel	Temporary data storage (buffer register)

Counters

DEFINITION
A **counter** is a sequential circuit that goes through a fixed sequence of states when clock pulses arrive. An *n*-bit counter has up to $\mathbf{2^n}$ states (a **mod-2^n** counter).

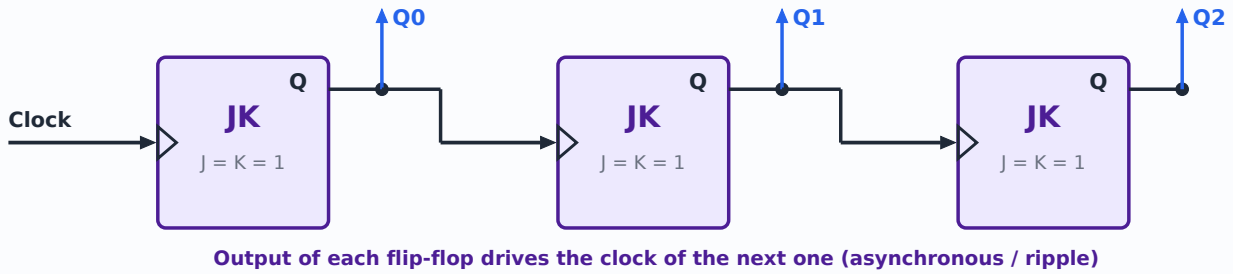
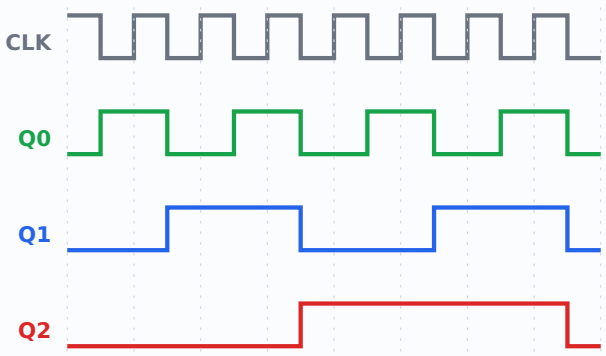


Fig 3.11 3-bit asynchronous (ripple) up-counter using JK flip-flops in toggle mode



Clock	Q2	Q1	Q0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Fig 3.12 Timing diagram and count sequence of the 3-bit counter (each stage divides the frequency by 2)

	Asynchronous (ripple) counter	Synchronous counter
Clock	Clock drives only the first FF; each FF clocks the next	**Same clock** to all flip-flops
Speed	Slower (delays add up)	Faster
Circuit	Simple, fewer gates	Needs extra gates; more complex

- **Up / down counters** count 0→7 or 7→0 (down counter takes the clock from Q′). **Mod-N counter:** resets after N states, e.g. a **decade (mod-10) counter** uses 4 flip-flops and clears at 1010.
- **Applications:** digital clocks and timers, frequency dividers, event counting, address generation in memories.

8 Quick Summary and Exam Corner

Unit III in one glance

- Binary (base 2), octal (8), hex (16). Group bits in 3s for octal and 4s for hex. 2’s complement = 1’s complement + 1.
- BCD = digit-by-digit 8421; Excess-3 = BCD + 3; Gray = one-bit change; Hamming (7,4) = 3 parity bits at 1, 2, 4 correct a single error.
- Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR; NAND and NOR are universal. De Morgan: $(AB)' = A'+B'$, $(A+B)' = A'B'$.
- Half adder: $S = A \oplus B$, $C = AB$. Full adder: $S = A \oplus B \oplus Cin$, $Cout = AB + Cin(A \oplus B)$.
- Sequential = combinational + memory. Flip-flops: SR, JK (toggle at 11), D ($Q^+ = D$), T (toggle).
- Shift registers: SISO, SIPO, PISO, PIPO. n flip-flops give a mod-2^n counter; ripple = asynchronous, others = synchronous.

Short answer questions (2 marks)

1. Convert $(45)_{10}$ to binary, and $(1011\ 1100)_2$ to hexadecimal.
2. What is a universal gate? Name two.
3. State De Morgan’s theorems.
4. Why is Gray code used in position encoders?
5. Write the Sum and Carry expressions of a half adder.
6. Differentiate combinational and sequential circuits.
7. What is a flip-flop? How many states does it have?
8. What is the use of a shift register?

Long answer questions (5 / 10 marks)

1. Explain BCD, Excess-3 and Gray codes with a table for the digits 0 to 9.
2. Explain the Hamming code. Encode 1011 using even parity and show how a single-bit error is corrected.
3. Draw the symbol, truth table and expression of all basic gates (NOT, OR, AND, NOR, NAND, XOR, XNOR).

4. Show that NAND and NOR gates are universal by realising NOT, AND and OR gates using each.
5. State and prove the basic theorems of Boolean algebra, including De Morgan's theorems, using truth tables.
6. Design a half adder and a full adder; give truth tables and logic diagrams.
7. Explain the SR, JK, D and T flip-flops with symbols and characteristic tables.
8. Explain a 4-bit shift register and a 3-bit ripple counter with diagrams and timing.

Numerical practice (with answers)

1. Convert $(110101)_2$ to decimal, octal and hex. *Ans: 53, 65_8, 35_{16}*
2. Find the 2's complement of 01100. *Ans: 10011 + 1 = 10100*
3. Convert binary 1100 to Gray code. *Ans: 1010*
4. Simplify Y = AB + AB′. *Ans: Y = A(B + B′) = A*
5. How many flip-flops are needed for a mod-12 counter? *Ans: 4 (since $2^3 = 8 < 12 \le 2^4 = 16$)*

Quick check: pick the right answer

1. The output of a 2-input XOR gate is 1 when the inputs are: (a) same (b) different (c) both 1. *Ans: (b)*
2. Which flip-flop toggles for J = K = 1? (a) SR (b) D (c) JK. *Ans: (c)*
3. A 3-bit counter has how many states? (a) 3 (b) 6 (c) 8. *Ans: (c)*
4. Excess-3 code of decimal 4 is: (a) 0100 (b) 0111 (c) 1000. *Ans: (b)*